Material Characterization Report

NanoVNA UTN Toolkit

2026-10-24 12:16:43

Measurement Details:

- **Sample:** Muestra5
- **Technique:** Open-ended coaxial probe (liquids)
- **Temperature:** 23.0 °C
- **Reference Liquids:** Distilled Water / Isopropyl Alcohol (IPA)
- **Probe:** 3 mm probe
- **Frequency Range:** 0.005–6.000 GHz, 1200 pts
- **Data sources:**
  - **Open:** kit:cal_sonda3mm_UBA_1200p
  - **Short:** kit:cal_sonda3mm_UBA_1200p (1 spike(s) replaced in the computation: 4885 MHz)
  - **Ref 1:** kit:cal_sonda3mm_UBA_1200p
  - **Ref 2:** kit:cal_sonda3mm_UBA_1200p
  - **DUT:** imported

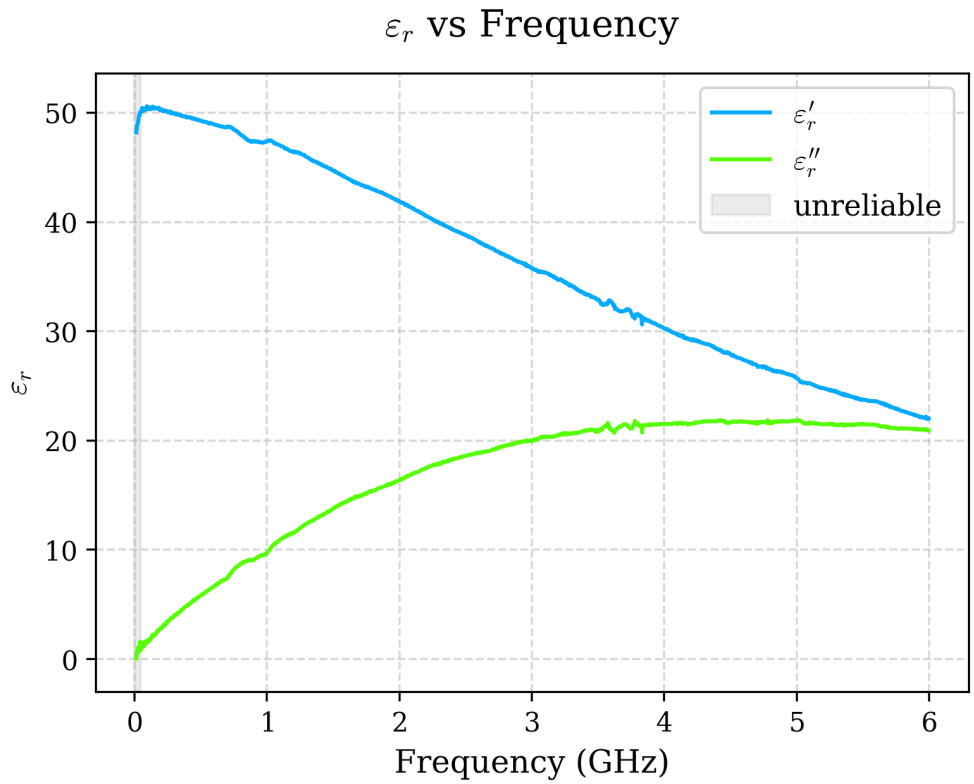
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1 Measurement Results

1.1 Complex Permittivity $\varepsilon_r(f)$



Points without a solution: 2 of 1200 (left blank in the plot and shown as — in the table).

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.005	—	—	—
0.010	—	—	—
0.015	48.1646	0.0113	0.0002
0.020	48.7854	0.5785	0.0119
0.025	48.9400	0.9843	0.0201
0.030	49.2624	0.9809	0.0199
0.035	49.7187	0.6287	0.0126
0.040	49.7794	0.9605	0.0193
0.045	50.0423	1.5698	0.0314
0.050	50.0394	0.8252	0.0165
0.055	50.0984	1.0969	0.0219
0.060	50.4161	1.0736	0.0213
0.065	50.1008	1.4674	0.0293
0.070	50.1816	1.3634	0.0272
0.075	50.1673	1.1318	0.0226
0.080	50.2159	1.5947	0.0318
0.085	50.3775	1.5161	0.0301

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.090	50.2583	1.4182	0.0282
0.095	50.5666	1.4691	0.0291
0.100	50.4885	1.5432	0.0306
0.105	50.3144	1.7411	0.0346
0.110	50.3798	1.6068	0.0319
0.115	50.4562	1.7987	0.0356
0.120	50.2774	1.7056	0.0339
0.125	50.4580	1.8393	0.0365
0.130	50.3348	2.0563	0.0409
0.135	50.4961	1.9861	0.0393
0.140	50.5311	2.2219	0.0440
0.145	50.3575	2.0831	0.0414
0.150	50.3269	2.1347	0.0424
0.155	50.4462	2.2008	0.0436
0.160	50.3547	2.2431	0.0445
0.165	50.3697	2.3490	0.0466
0.170	50.3645	2.3909	0.0475
0.175	50.4101	2.5919	0.0514
0.180	50.3798	2.5384	0.0504
0.185	50.4297	2.5970	0.0515
0.190	50.2845	2.7130	0.0540
0.195	50.1893	2.7724	0.0552
0.200	50.1947	2.8032	0.0558
0.205	50.2962	2.7679	0.0550
0.210	50.2276	2.9163	0.0581
0.215	50.1154	2.9089	0.0580
0.220	50.1719	3.1181	0.0621
0.225	50.1463	3.0112	0.0600
0.230	50.1361	3.0595	0.0610
0.235	50.1797	3.1177	0.0621
0.240	50.1473	3.2765	0.0653
0.245	50.1193	3.3989	0.0678
0.250	50.0236	3.4535	0.0690
0.255	50.0960	3.3947	0.0678
0.260	50.0813	3.4698	0.0693
0.265	50.0065	3.5254	0.0705
0.270	49.9636	3.6009	0.0721
0.275	50.0800	3.7316	0.0745
0.280	49.8999	3.6758	0.0737
0.285	49.9755	3.6946	0.0739
0.290	49.9348	3.7749	0.0756
0.295	49.9318	3.8565	0.0772
0.300	49.8877	3.8700	0.0776
0.305	49.8660	3.9986	0.0802
0.310	49.8169	4.0292	0.0809
0.315	49.9292	4.0083	0.0803

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.320	49.8664	4.0168	0.0806
0.325	49.7978	4.1926	0.0842
0.330	49.7793	4.1798	0.0840
0.335	49.7958	4.2207	0.0848
0.340	49.7687	4.2643	0.0857
0.345	49.7968	4.2665	0.0857
0.350	49.7387	4.4069	0.0886
0.355	49.6728	4.4935	0.0905
0.360	49.7032	4.4922	0.0904
0.365	49.6776	4.5276	0.0911
0.370	49.6444	4.5689	0.0920
0.375	49.6254	4.6749	0.0942
0.380	49.6496	4.7045	0.0948
0.385	49.6364	4.7576	0.0958
0.390	49.6368	4.8119	0.0969
0.395	49.5377	4.7991	0.0969
0.400	49.5331	4.8406	0.0977
0.405	49.5600	4.9592	0.1001
0.410	49.5728	4.9166	0.0992
0.415	49.5354	5.0068	0.1011
0.420	49.5419	5.0555	0.1020
0.425	49.5299	5.0994	0.1030
0.430	49.5209	5.1802	0.1046
0.435	49.4239	5.1831	0.1049
0.440	49.4770	5.2483	0.1061
0.445	49.4172	5.3659	0.1086
0.450	49.4337	5.3916	0.1091
0.455	49.3835	5.3513	0.1084
0.460	49.4126	5.4567	0.1104
0.465	49.3827	5.4675	0.1107
0.470	49.3464	5.5522	0.1125
0.475	49.3403	5.5749	0.1130
0.480	49.3408	5.6173	0.1138
0.485	49.2766	5.6398	0.1145
0.490	49.2702	5.6994	0.1157
0.495	49.2571	5.7620	0.1170
0.500	49.2316	5.7742	0.1173
0.505	49.2413	5.8014	0.1178
0.510	49.2024	5.8384	0.1187
0.515	49.1617	5.9322	0.1207
0.520	49.1835	5.8996	0.1200
0.525	49.1607	5.9750	0.1215
0.530	49.1465	6.0415	0.1229
0.535	49.1238	6.0875	0.1239
0.540	49.1334	6.1330	0.1248
0.545	49.1122	6.1780	0.1258

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.550	49.0748	6.1727	0.1258
0.555	49.0649	6.2343	0.1271
0.560	48.9840	6.2696	0.1280
0.565	49.0111	6.3201	0.1290
0.570	48.9912	6.3979	0.1306
0.575	49.0390	6.4148	0.1308
0.580	49.0029	6.4709	0.1321
0.585	48.9585	6.4707	0.1322
0.590	48.9515	6.4997	0.1328
0.595	48.9301	6.6059	0.1350
0.600	48.9421	6.6430	0.1357
0.605	48.8863	6.6748	0.1365
0.610	48.8641	6.7130	0.1374
0.615	48.8597	6.7687	0.1385
0.620	48.8370	6.7622	0.1385
0.625	48.8220	6.8512	0.1403
0.630	48.8201	6.8578	0.1405
0.635	48.7846	6.9434	0.1423
0.640	48.7883	6.9619	0.1427
0.645	48.7276	6.9901	0.1435
0.650	48.7269	7.0473	0.1446
0.655	48.7193	7.1110	0.1460
0.660	48.7085	7.0958	0.1457
0.665	48.7246	7.1424	0.1466
0.670	48.6865	7.2092	0.1481
0.675	48.6769	7.1909	0.1477
0.680	48.6351	7.2511	0.1491
0.685	48.6424	7.2611	0.1493
0.690	48.6710	7.2639	0.1492
0.695	48.6487	7.3050	0.1502
0.700	48.6337	7.3802	0.1518
0.705	48.7143	7.4929	0.1538
0.710	48.6981	7.5041	0.1541
0.715	48.6855	7.6204	0.1565
0.720	48.6927	7.7271	0.1587
0.725	48.6560	7.7780	0.1599
0.730	48.6457	7.8182	0.1607
0.735	48.5666	7.9244	0.1632
0.740	48.5584	8.0175	0.1651
0.745	48.5378	8.1013	0.1669
0.750	48.4347	8.1589	0.1685
0.755	48.3929	8.1908	0.1693
0.760	48.3909	8.2695	0.1709
0.765	48.3524	8.3437	0.1726
0.770	48.2958	8.3987	0.1739
0.775	48.3021	8.4004	0.1739

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.780	48.2125	8.4928	0.1762
0.785	48.1470	8.5280	0.1771
0.790	48.1209	8.5255	0.1772
0.795	48.0583	8.6067	0.1791
0.800	48.0000	8.6297	0.1798
0.805	47.9363	8.7167	0.1818
0.810	47.9062	8.7383	0.1824
0.815	47.8976	8.8109	0.1840
0.820	47.8181	8.8367	0.1848
0.825	47.7553	8.8023	0.1843
0.830	47.7484	8.8654	0.1857
0.835	47.6710	8.8956	0.1866
0.840	47.6093	8.8827	0.1866
0.845	47.6197	8.9072	0.1870
0.850	47.5310	8.9403	0.1881
0.855	47.4949	9.0042	0.1896
0.860	47.4276	8.9697	0.1891
0.865	47.3862	9.0572	0.1911
0.870	47.4055	9.0589	0.1911
0.875	47.3982	9.0027	0.1899
0.880	47.3117	8.9723	0.1896
0.885	47.2939	9.0018	0.1903
0.890	47.3515	9.0472	0.1911
0.895	47.3121	9.0197	0.1906
0.900	47.3789	9.0057	0.1901
0.905	47.3920	9.0961	0.1919
0.910	47.3632	9.1321	0.1928
0.915	47.3160	9.1694	0.1938
0.920	47.3149	9.1653	0.1937
0.925	47.3583	9.2737	0.1958
0.930	47.3418	9.2805	0.1960
0.935	47.2883	9.3155	0.1970
0.940	47.3243	9.3739	0.1981
0.945	47.2798	9.3937	0.1987
0.950	47.2716	9.3790	0.1984
0.955	47.2498	9.4483	0.2000
0.960	47.2305	9.4475	0.2000
0.965	47.2460	9.5028	0.2011
0.970	47.2536	9.4904	0.2008
0.975	47.2303	9.4868	0.2009
0.980	47.2776	9.5163	0.2013
0.985	47.3116	9.5223	0.2013
0.990	47.2894	9.5777	0.2025
0.995	47.3340	9.6104	0.2030
1.000	47.3547	9.6108	0.2030
1.005	47.3861	9.7431	0.2056

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.010	47.4170	9.8404	0.2075
1.015	47.3735	9.8778	0.2085
1.020	47.4175	9.9723	0.2103
1.025	47.4550	10.0445	0.2117
1.030	47.4099	10.1259	0.2136
1.035	47.4038	10.1977	0.2151
1.040	47.3646	10.2842	0.2171
1.045	47.3122	10.3478	0.2187
1.050	47.2743	10.4305	0.2206
1.055	47.1710	10.4300	0.2211
1.060	47.1987	10.5439	0.2234
1.065	47.1364	10.5105	0.2230
1.070	47.1532	10.6032	0.2249
1.075	47.0912	10.6781	0.2268
1.080	47.0765	10.6743	0.2267
1.085	47.0493	10.7029	0.2275
1.090	47.0482	10.7809	0.2291
1.095	46.9955	10.8442	0.2308
1.100	46.9866	10.9004	0.2320
1.105	46.9517	10.9052	0.2323
1.110	46.8902	10.9785	0.2341
1.115	46.8921	11.0259	0.2351
1.120	46.8794	11.0436	0.2356
1.125	46.8453	11.0748	0.2364
1.130	46.7869	11.1522	0.2384
1.135	46.7136	11.1570	0.2388
1.140	46.6926	11.1352	0.2385
1.145	46.6744	11.2375	0.2408
1.150	46.6961	11.2208	0.2403
1.155	46.6451	11.3111	0.2425
1.160	46.5852	11.2898	0.2423
1.165	46.5821	11.3703	0.2441
1.170	46.5452	11.3483	0.2438
1.175	46.4603	11.3984	0.2453
1.180	46.4810	11.4580	0.2465
1.185	46.4679	11.4401	0.2462
1.190	46.4203	11.4536	0.2467
1.195	46.4188	11.5076	0.2479
1.200	46.4281	11.5547	0.2489
1.205	46.3323	11.5219	0.2487
1.210	46.3845	11.5844	0.2497
1.215	46.3397	11.6342	0.2511
1.220	46.3851	11.6361	0.2509
1.225	46.3733	11.6930	0.2521
1.230	46.3652	11.7734	0.2539
1.235	46.2943	11.7957	0.2548

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.240	46.3403	11.8406	0.2555
1.245	46.3173	11.8943	0.2568
1.250	46.2315	11.9355	0.2582
1.255	46.2062	11.9780	0.2592
1.260	46.2105	12.0063	0.2598
1.265	46.1540	12.0861	0.2619
1.270	46.1416	12.1094	0.2624
1.275	46.1214	12.1634	0.2637
1.280	46.0879	12.2120	0.2650
1.285	46.1066	12.2843	0.2664
1.290	46.0095	12.2595	0.2665
1.295	45.9506	12.3238	0.2682
1.300	45.9330	12.3679	0.2693
1.305	45.9081	12.3961	0.2700
1.310	45.8687	12.4533	0.2715
1.315	45.8476	12.4837	0.2723
1.320	45.7714	12.4821	0.2727
1.325	45.7302	12.5323	0.2740
1.330	45.6871	12.5680	0.2751
1.335	45.6538	12.5922	0.2758
1.340	45.6656	12.6214	0.2764
1.345	45.5809	12.6403	0.2773
1.350	45.5577	12.7058	0.2789
1.355	45.5162	12.6666	0.2783
1.360	45.5477	12.7524	0.2800
1.365	45.5040	12.7892	0.2811
1.370	45.4648	12.8212	0.2820
1.375	45.4861	12.8610	0.2827
1.380	45.4091	12.8918	0.2839
1.385	45.3841	12.9477	0.2853
1.390	45.3361	12.9562	0.2858
1.395	45.3178	13.0361	0.2877
1.400	45.3278	13.0510	0.2879
1.405	45.2360	13.0800	0.2891
1.410	45.2570	13.1220	0.2899
1.415	45.1858	13.1027	0.2900
1.420	45.1965	13.1738	0.2915
1.425	45.1512	13.1448	0.2911
1.430	45.0744	13.2267	0.2934
1.435	45.0672	13.2406	0.2938
1.440	45.0425	13.2923	0.2951
1.445	45.0188	13.3216	0.2959
1.450	45.0388	13.3738	0.2969
1.455	44.9763	13.4166	0.2983
1.460	44.9449	13.4987	0.3003
1.465	44.9127	13.5073	0.3007

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.470	44.8666	13.5122	0.3012
1.475	44.8328	13.4703	0.3005
1.480	44.8465	13.5781	0.3028
1.485	44.7861	13.6100	0.3039
1.490	44.7897	13.6485	0.3047
1.495	44.7251	13.6801	0.3059
1.500	44.6841	13.7487	0.3077
1.505	44.6548	13.7834	0.3087
1.510	44.5967	13.7641	0.3086
1.515	44.6407	13.8854	0.3110
1.520	44.5425	13.9057	0.3122
1.525	44.5316	13.9491	0.3132
1.530	44.5082	13.9442	0.3133
1.535	44.4761	13.9719	0.3141
1.540	44.4237	14.0027	0.3152
1.545	44.3802	14.0850	0.3174
1.550	44.3920	14.0912	0.3174
1.555	44.3383	14.0926	0.3178
1.560	44.2955	14.1605	0.3197
1.565	44.2452	14.2177	0.3213
1.570	44.2078	14.2105	0.3214
1.575	44.1830	14.2746	0.3231
1.580	44.1992	14.2610	0.3227
1.585	44.1186	14.2814	0.3237
1.590	44.0782	14.3354	0.3252
1.595	44.0591	14.3577	0.3259
1.600	44.0227	14.3852	0.3268
1.605	44.0133	14.4203	0.3276
1.610	43.9634	14.4036	0.3276
1.615	43.9299	14.4706	0.3294
1.620	43.9059	14.5497	0.3314
1.625	43.8812	14.5834	0.3323
1.630	43.8193	14.5263	0.3315
1.635	43.8035	14.6208	0.3338
1.640	43.7976	14.6340	0.3341
1.645	43.7342	14.6560	0.3351
1.650	43.6708	14.6333	0.3351
1.655	43.6598	14.6679	0.3360
1.660	43.5988	14.7581	0.3385
1.665	43.6534	14.8347	0.3398
1.670	43.5782	14.8298	0.3403
1.675	43.5215	14.8203	0.3405
1.680	43.5610	14.8484	0.3409
1.685	43.5002	14.7721	0.3396
1.690	43.4892	14.8730	0.3420
1.695	43.4773	14.8628	0.3419

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.700	43.4352	14.8901	0.3428
1.705	43.3865	14.9259	0.3440
1.710	43.3899	14.9280	0.3440
1.715	43.3795	14.8887	0.3432
1.720	43.3097	14.9829	0.3459
1.725	43.2938	14.9714	0.3458
1.730	43.3304	15.0275	0.3468
1.735	43.2610	15.0662	0.3483
1.740	43.2231	15.0870	0.3491
1.745	43.1601	15.1427	0.3508
1.750	43.1949	15.0728	0.3489
1.755	43.2129	15.1723	0.3511
1.760	43.1737	15.1735	0.3515
1.765	43.1883	15.2252	0.3525
1.770	43.1130	15.2411	0.3535
1.775	43.1119	15.3313	0.3556
1.780	43.0849	15.2786	0.3546
1.785	43.0106	15.3263	0.3563
1.790	43.0107	15.3056	0.3559
1.795	43.0127	15.3533	0.3569
1.800	42.9858	15.3351	0.3567
1.805	42.9316	15.3776	0.3582
1.810	42.9295	15.3988	0.3587
1.815	42.9112	15.4154	0.3592
1.820	42.8967	15.4460	0.3601
1.825	42.8759	15.4786	0.3610
1.830	42.8012	15.5070	0.3623
1.835	42.7962	15.5479	0.3633
1.840	42.7668	15.5806	0.3643
1.845	42.7494	15.5978	0.3649
1.850	42.7269	15.6162	0.3655
1.855	42.7106	15.6282	0.3659
1.860	42.6709	15.6206	0.3661
1.865	42.6900	15.6740	0.3672
1.870	42.6758	15.7075	0.3681
1.875	42.6190	15.7431	0.3694
1.880	42.6052	15.7387	0.3694
1.885	42.5316	15.7518	0.3704
1.890	42.5255	15.7730	0.3709
1.895	42.4951	15.8150	0.3722
1.900	42.4702	15.8897	0.3741
1.905	42.4465	15.8705	0.3739
1.910	42.4127	15.8848	0.3745
1.915	42.3684	15.9452	0.3763
1.920	42.3542	15.9329	0.3762
1.925	42.3384	15.9873	0.3776

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.930	42.3122	15.9465	0.3769
1.935	42.2937	16.0095	0.3785
1.940	42.2270	15.9501	0.3777
1.945	42.1753	16.0799	0.3813
1.950	42.1518	16.0808	0.3815
1.955	42.1303	16.1200	0.3826
1.960	42.1259	16.1101	0.3824
1.965	42.0578	16.1370	0.3837
1.970	42.0587	16.1555	0.3841
1.975	42.0299	16.2458	0.3865
1.980	41.9642	16.2065	0.3862
1.985	41.9860	16.2106	0.3861
1.990	41.9303	16.2629	0.3879
1.995	41.8961	16.3022	0.3891
2.000	41.8267	16.3506	0.3909
2.005	41.8574	16.3463	0.3905
2.010	41.8014	16.4465	0.3934
2.015	41.7481	16.3765	0.3923
2.020	41.7115	16.4220	0.3937
2.025	41.7118	16.4844	0.3952
2.030	41.7311	16.4782	0.3949
2.035	41.6736	16.5412	0.3969
2.040	41.6483	16.5606	0.3976
2.045	41.5312	16.5583	0.3987
2.050	41.5817	16.6857	0.4013
2.055	41.5636	16.6739	0.4012
2.060	41.4779	16.7106	0.4029
2.065	41.5210	16.7322	0.4030
2.070	41.4243	16.7260	0.4038
2.075	41.3658	16.7385	0.4046
2.080	41.3578	16.7928	0.4060
2.085	41.3307	16.8300	0.4072
2.090	41.3156	16.8504	0.4078
2.095	41.2737	16.8612	0.4085
2.100	41.2161	16.9628	0.4116
2.105	41.2034	16.9439	0.4112
2.110	41.2060	16.9838	0.4122
2.115	41.1259	16.9990	0.4133
2.120	41.0767	16.9841	0.4135
2.125	41.0312	17.0472	0.4155
2.130	41.0195	17.0522	0.4157
2.135	41.0085	17.0901	0.4167
2.140	40.9838	17.1173	0.4177
2.145	40.9233	17.1303	0.4186
2.150	40.9426	17.1415	0.4187
2.155	40.8703	17.1743	0.4202

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.160	40.8138	17.2178	0.4219
2.165	40.8320	17.2490	0.4224
2.170	40.7570	17.2693	0.4237
2.175	40.7536	17.3215	0.4250
2.180	40.7110	17.3344	0.4258
2.185	40.6961	17.3654	0.4267
2.190	40.6470	17.3859	0.4277
2.195	40.5443	17.4045	0.4293
2.200	40.5354	17.4395	0.4302
2.205	40.5209	17.4144	0.4298
2.210	40.5092	17.5151	0.4324
2.215	40.4598	17.5021	0.4326
2.220	40.4415	17.5322	0.4335
2.225	40.3823	17.5926	0.4357
2.230	40.3552	17.6096	0.4364
2.235	40.3437	17.6270	0.4369
2.240	40.3168	17.6084	0.4368
2.245	40.2454	17.6839	0.4394
2.250	40.2343	17.6745	0.4393
2.255	40.2195	17.6898	0.4398
2.260	40.1489	17.7475	0.4420
2.265	40.1311	17.7719	0.4428
2.270	40.1020	17.7168	0.4418
2.275	40.0229	17.7582	0.4437
2.280	40.0203	17.7621	0.4438
2.285	40.0120	17.8461	0.4460
2.290	39.9600	17.8295	0.4462
2.295	39.9878	17.8113	0.4454
2.300	39.9209	17.7883	0.4456
2.305	39.8482	17.9077	0.4494
2.310	39.8539	17.8147	0.4470
2.315	39.8654	17.8978	0.4490
2.320	39.8030	17.9354	0.4506
2.325	39.7343	17.8883	0.4502
2.330	39.7461	18.0413	0.4539
2.335	39.6931	18.0060	0.4536
2.340	39.7145	17.9653	0.4524
2.345	39.6175	18.0383	0.4553
2.350	39.5970	18.0691	0.4563
2.355	39.5640	18.0121	0.4553
2.360	39.5565	18.0997	0.4576
2.365	39.5281	18.1341	0.4588
2.370	39.5026	18.1022	0.4583
2.375	39.4886	18.0796	0.4578
2.380	39.4539	18.1726	0.4606
2.385	39.3941	18.1870	0.4617

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.390	39.4115	18.2063	0.4620
2.395	39.3368	18.2673	0.4644
2.400	39.2894	18.2623	0.4648
2.405	39.2783	18.2125	0.4637
2.410	39.2682	18.2646	0.4651
2.415	39.2118	18.2806	0.4662
2.420	39.1956	18.3452	0.4680
2.425	39.1595	18.3013	0.4674
2.430	39.1386	18.3110	0.4678
2.435	39.1514	18.3284	0.4681
2.440	39.0961	18.3584	0.4696
2.445	39.0754	18.3762	0.4703
2.450	39.0785	18.4005	0.4709
2.455	39.0098	18.3964	0.4716
2.460	38.9985	18.4189	0.4723
2.465	38.9903	18.4755	0.4738
2.470	38.9262	18.4445	0.4738
2.475	38.9081	18.4665	0.4746
2.480	38.8712	18.4945	0.4758
2.485	38.8778	18.5397	0.4769
2.490	38.8370	18.5314	0.4772
2.495	38.7867	18.5678	0.4787
2.500	38.8113	18.5653	0.4783
2.505	38.7592	18.5695	0.4791
2.510	38.7426	18.5962	0.4800
2.515	38.7038	18.6048	0.4807
2.520	38.6828	18.6496	0.4821
2.525	38.5975	18.6457	0.4831
2.530	38.5968	18.6657	0.4836
2.535	38.6213	18.6803	0.4837
2.540	38.5654	18.6687	0.4841
2.545	38.5337	18.6777	0.4847
2.550	38.4894	18.7129	0.4862
2.555	38.4994	18.7133	0.4861
2.560	38.4405	18.7519	0.4878
2.565	38.4033	18.7557	0.4884
2.570	38.3855	18.7617	0.4888
2.575	38.3562	18.7849	0.4897
2.580	38.3070	18.7732	0.4901
2.585	38.2695	18.8503	0.4926
2.590	38.2409	18.7916	0.4914
2.595	38.1996	18.8454	0.4933
2.600	38.1493	18.8337	0.4937
2.605	38.0815	18.8342	0.4946
2.610	38.1378	18.8589	0.4945
2.615	38.0822	18.8335	0.4945

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.620	38.0117	18.8857	0.4968
2.625	37.9741	18.8888	0.4974
2.630	37.9779	18.9332	0.4985
2.635	37.9652	18.9096	0.4981
2.640	37.8782	18.9391	0.5000
2.645	37.8884	18.9436	0.5000
2.650	37.8056	18.9721	0.5018
2.655	37.8172	18.9450	0.5010
2.660	37.7624	18.9859	0.5028
2.665	37.7992	18.9823	0.5022
2.670	37.7390	18.9531	0.5022
2.675	37.6923	18.9930	0.5039
2.680	37.6896	19.0010	0.5041
2.685	37.6245	19.0303	0.5058
2.690	37.5956	19.0402	0.5064
2.695	37.5856	19.0702	0.5074
2.700	37.5570	19.0961	0.5085
2.705	37.5452	19.1018	0.5088
2.710	37.5040	19.1854	0.5116
2.715	37.4977	19.1304	0.5102
2.720	37.4452	19.1846	0.5123
2.725	37.3619	19.1911	0.5137
2.730	37.3153	19.2216	0.5151
2.735	37.3073	19.2853	0.5169
2.740	37.3031	19.2038	0.5148
2.745	37.2639	19.3262	0.5186
2.750	37.2256	19.3337	0.5194
2.755	37.1717	19.3508	0.5206
2.760	37.2002	19.3737	0.5208
2.765	37.1175	19.3506	0.5213
2.770	37.1355	19.4362	0.5234
2.775	37.0511	19.3908	0.5234
2.780	37.0446	19.4067	0.5239
2.785	37.0185	19.4528	0.5255
2.790	37.0133	19.4206	0.5247
2.795	36.9672	19.4948	0.5274
2.800	36.9303	19.4889	0.5277
2.805	36.9313	19.5184	0.5285
2.810	36.8349	19.5116	0.5297
2.815	36.8450	19.5558	0.5308
2.820	36.8381	19.5468	0.5306
2.825	36.7830	19.5368	0.5311
2.830	36.7717	19.5884	0.5327
2.835	36.7648	19.5920	0.5329
2.840	36.6911	19.6379	0.5352
2.845	36.6734	19.6529	0.5359

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.850	36.6664	19.5813	0.5340
2.855	36.6449	19.6680	0.5367
2.860	36.5808	19.6421	0.5370
2.865	36.5175	19.6874	0.5391
2.870	36.4888	19.6914	0.5397
2.875	36.4832	19.6818	0.5395
2.880	36.4483	19.7508	0.5419
2.885	36.4466	19.7411	0.5416
2.890	36.3621	19.8209	0.5451
2.895	36.3505	19.7561	0.5435
2.900	36.3527	19.7831	0.5442
2.905	36.3123	19.8455	0.5465
2.910	36.2988	19.8748	0.5475
2.915	36.2640	19.8497	0.5474
2.920	36.2600	19.8683	0.5479
2.925	36.2598	19.8228	0.5467
2.930	36.2103	19.8562	0.5484
2.935	36.1667	19.8720	0.5495
2.940	36.1766	19.8945	0.5499
2.945	36.1262	19.9176	0.5513
2.950	36.0667	19.9044	0.5519
2.955	36.1594	19.8876	0.5500
2.960	35.9404	19.9908	0.5562
2.965	36.0411	19.8966	0.5521
2.970	35.9333	19.8790	0.5532
2.975	35.9249	19.9053	0.5541
2.980	35.9538	19.9453	0.5547
2.985	35.8677	19.9380	0.5559
2.990	35.8244	20.0050	0.5584
2.995	35.8112	19.9289	0.5565
3.000	35.7222	20.0016	0.5599
3.005	35.7667	19.9894	0.5589
3.010	35.7133	19.9214	0.5578
3.015	35.6728	19.9623	0.5596
3.020	35.6090	20.0889	0.5642
3.025	35.5854	20.0325	0.5629
3.030	35.5996	20.0141	0.5622
3.035	35.6127	20.0916	0.5642
3.040	35.5224	20.1055	0.5660
3.045	35.4862	20.1020	0.5665
3.050	35.3923	20.0908	0.5677
3.055	35.4577	20.2063	0.5699
3.060	35.4295	20.2349	0.5711
3.065	35.4759	20.1745	0.5687
3.070	35.3918	20.2885	0.5733
3.075	35.4200	20.2891	0.5728

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.080	35.4143	20.3257	0.5739
3.085	35.3498	20.2582	0.5731
3.090	35.3544	20.3015	0.5742
3.095	35.3695	20.3231	0.5746
3.100	35.3774	20.3052	0.5740
3.105	35.3166	20.2641	0.5738
3.110	35.3331	20.3397	0.5757
3.115	35.3093	20.3067	0.5751
3.120	35.2780	20.3424	0.5766
3.125	35.2357	20.3101	0.5764
3.130	35.1981	20.3173	0.5772
3.135	35.1427	20.3213	0.5782
3.140	35.1595	20.3190	0.5779
3.145	35.0732	20.2948	0.5786
3.150	35.0816	20.3142	0.5791
3.155	35.0284	20.3409	0.5807
3.160	35.0367	20.3285	0.5802
3.165	34.9369	20.3587	0.5827
3.170	34.9859	20.3098	0.5805
3.175	34.8701	20.3235	0.5828
3.180	34.8823	20.3531	0.5835
3.185	34.8143	20.4080	0.5862
3.190	34.8090	20.4068	0.5863
3.195	34.7438	20.4436	0.5884
3.200	34.6793	20.4504	0.5897
3.205	34.7026	20.5108	0.5910
3.210	34.6687	20.4976	0.5912
3.215	34.6768	20.4756	0.5905
3.220	34.7290	20.5254	0.5910
3.225	34.6329	20.6024	0.5949
3.230	34.6313	20.4836	0.5915
3.235	34.6077	20.5800	0.5947
3.240	34.5258	20.5698	0.5958
3.245	34.5456	20.5270	0.5942
3.250	34.4330	20.6770	0.6005
3.255	34.3856	20.5930	0.5989
3.260	34.4557	20.5902	0.5976
3.265	34.4126	20.6521	0.6001
3.270	34.3205	20.5185	0.5979
3.275	34.3033	20.5767	0.5998
3.280	34.3194	20.5995	0.6002
3.285	34.1477	20.6422	0.6045
3.290	34.2267	20.5588	0.6007
3.295	34.1483	20.5811	0.6027
3.300	34.1930	20.5856	0.6020
3.305	34.1308	20.6434	0.6048

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.310	34.0663	20.6220	0.6054
3.315	34.0128	20.6106	0.6060
3.320	34.0325	20.6673	0.6073
3.325	34.0350	20.6403	0.6064
3.330	33.9517	20.6901	0.6094
3.335	33.8352	20.6411	0.6100
3.340	33.8622	20.6818	0.6108
3.345	33.7945	20.6408	0.6108
3.350	33.7270	20.6858	0.6133
3.355	33.7539	20.6667	0.6123
3.360	33.6659	20.7379	0.6160
3.365	33.6397	20.7390	0.6165
3.370	33.5856	20.7616	0.6182
3.375	33.6288	20.7915	0.6183
3.380	33.5916	20.8032	0.6193
3.385	33.5385	20.8204	0.6208
3.390	33.4961	20.7979	0.6209
3.395	33.4936	20.8183	0.6216
3.400	33.3836	20.8799	0.6255
3.405	33.3876	20.8579	0.6247
3.410	33.3802	20.9163	0.6266
3.415	33.2986	20.9076	0.6279
3.420	33.3197	20.8836	0.6268
3.425	33.3466	20.8885	0.6264
3.430	33.3196	20.9187	0.6278
3.435	33.3007	20.9813	0.6301
3.440	33.2742	20.8992	0.6281
3.445	33.1986	20.9434	0.6308
3.450	33.1046	20.9350	0.6324
3.455	33.1620	20.9833	0.6327
3.460	33.1042	20.9648	0.6333
3.465	33.0744	20.9588	0.6337
3.470	33.0732	20.9431	0.6332
3.475	33.0629	20.8923	0.6319
3.480	33.0023	20.8580	0.6320
3.485	32.9986	20.7452	0.6287
3.490	32.9112	20.9500	0.6366
3.495	32.8640	20.9476	0.6374
3.500	32.8451	20.8696	0.6354
3.505	32.8935	20.9300	0.6363
3.510	32.8005	20.8087	0.6344
3.515	32.7167	20.9269	0.6396
3.520	32.6482	20.8865	0.6397
3.525	32.4570	21.0235	0.6477
3.530	32.5287	21.0693	0.6477
3.535	32.4420	20.9906	0.6470

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.540	32.3634	21.1131	0.6524
3.545	32.4706	21.2119	0.6533
3.550	32.3861	21.2622	0.6565
3.555	32.4300	21.2415	0.6550
3.560	32.4769	21.3242	0.6566
3.565	32.5370	21.2975	0.6546
3.570	32.4267	21.3913	0.6597
3.575	32.4771	21.5851	0.6646
3.580	32.6512	21.5028	0.6586
3.585	32.8073	21.2978	0.6492
3.590	32.7812	21.1066	0.6439
3.595	32.7717	21.0073	0.6410
3.600	32.6910	21.0783	0.6448
3.605	32.6684	20.8882	0.6394
3.610	32.5782	20.8764	0.6408
3.615	32.4130	20.8787	0.6441
3.620	32.3013	20.8458	0.6454
3.625	32.3717	20.6767	0.6387
3.630	31.9977	20.8228	0.6508
3.635	32.0338	20.9399	0.6537
3.640	32.0638	21.0172	0.6555
3.645	31.9941	21.0803	0.6589
3.650	31.9035	21.1167	0.6619
3.655	31.8637	21.1001	0.6622
3.660	31.8542	21.2231	0.6663
3.665	31.8526	21.1827	0.6650
3.670	31.7844	21.2637	0.6690
3.675	31.7742	21.2310	0.6682
3.680	31.7955	21.2878	0.6695
3.685	31.7930	21.3760	0.6724
3.690	31.7919	21.3704	0.6722
3.695	31.8101	21.4353	0.6739
3.700	31.8490	21.4495	0.6735
3.705	31.8852	21.4361	0.6723
3.710	31.8456	21.4438	0.6734
3.715	31.9272	21.4418	0.6716
3.720	31.9986	21.3594	0.6675
3.725	31.9633	21.3855	0.6691
3.730	31.9918	21.3185	0.6664
3.735	31.9238	21.1945	0.6639
3.740	31.9321	21.1414	0.6621
3.745	31.8616	21.0777	0.6615
3.750	31.7202	21.0470	0.6635
3.755	31.6039	21.0484	0.6660
3.760	31.4486	21.1035	0.6710
3.765	31.3059	21.2308	0.6782

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.770	31.3453	21.3754	0.6819
3.775	31.3638	21.4337	0.6834
3.780	31.1270	21.7643	0.6992
3.785	31.4278	21.5991	0.6873
3.790	31.5291	21.4304	0.6797
3.795	31.5127	21.4422	0.6804
3.800	31.5072	21.5541	0.6841
3.805	31.5455	21.4043	0.6785
3.810	31.4370	21.4049	0.6809
3.815	31.3521	21.2900	0.6791
3.820	31.4107	21.3792	0.6806
3.825	31.3192	21.2895	0.6798
3.830	31.3218	21.3527	0.6817
3.835	30.5867	20.6903	0.6764
3.840	31.2048	21.3365	0.6838
3.845	31.1693	21.3661	0.6855
3.850	31.1565	21.3476	0.6852
3.855	31.1001	21.3929	0.6879
3.860	31.0478	21.3475	0.6876
3.865	30.9537	21.3838	0.6908
3.870	31.0479	21.3829	0.6887
3.875	30.9616	21.3926	0.6909
3.880	30.9905	21.3766	0.6898
3.885	30.9042	21.4182	0.6931
3.890	30.9339	21.4532	0.6935
3.895	30.9131	21.3600	0.6910
3.900	30.8469	21.3808	0.6931
3.905	30.7726	21.4644	0.6975
3.910	30.7995	21.4154	0.6953
3.915	30.7647	21.4198	0.6962
3.920	30.7367	21.4792	0.6988
3.925	30.6856	21.4233	0.6982
3.930	30.6686	21.4286	0.6987
3.935	30.6816	21.4196	0.6981
3.940	30.6147	21.4490	0.7006
3.945	30.5916	21.4129	0.7000
3.950	30.5398	21.4287	0.7017
3.955	30.5097	21.4616	0.7034
3.960	30.4892	21.4338	0.7030
3.965	30.4909	21.4396	0.7031
3.970	30.4366	21.4787	0.7057
3.975	30.4007	21.4711	0.7063
3.980	30.3772	21.4547	0.7063
3.985	30.3654	21.4933	0.7078
3.990	30.3527	21.4828	0.7078
3.995	30.2908	21.4592	0.7084

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.000	30.2580	21.4810	0.7099
4.005	30.2227	21.4825	0.7108
4.010	30.1827	21.4881	0.7119
4.015	30.1836	21.5231	0.7131
4.020	30.1483	21.4871	0.7127
4.025	30.1808	21.4443	0.7105
4.030	30.0689	21.4819	0.7144
4.035	30.0387	21.5173	0.7163
4.040	30.0580	21.4976	0.7152
4.045	30.0312	21.4733	0.7150
4.050	29.9904	21.4653	0.7157
4.055	29.9438	21.4672	0.7169
4.060	29.8937	21.4969	0.7191
4.065	29.8530	21.5037	0.7203
4.070	29.8656	21.5374	0.7211
4.075	29.8198	21.5088	0.7213
4.080	29.7980	21.5652	0.7237
4.085	29.7745	21.5366	0.7233
4.090	29.7699	21.5294	0.7232
4.095	29.6967	21.5834	0.7268
4.100	29.7296	21.5893	0.7262
4.105	29.6298	21.4514	0.7240
4.110	29.6807	21.5727	0.7268
4.115	29.6086	21.5531	0.7279
4.120	29.5560	21.5654	0.7296
4.125	29.6199	21.5515	0.7276
4.130	29.6143	21.6103	0.7297
4.135	29.4938	21.5450	0.7305
4.140	29.4905	21.4885	0.7287
4.145	29.5781	21.4872	0.7265
4.150	29.3863	21.5547	0.7335
4.155	29.3522	21.5721	0.7349
4.160	29.4041	21.5564	0.7331
4.165	29.3152	21.5058	0.7336
4.170	29.2901	21.5024	0.7341
4.175	29.3308	21.4465	0.7312
4.180	29.3318	21.5038	0.7331
4.185	29.2759	21.4365	0.7322
4.190	29.2159	21.4532	0.7343
4.195	29.2080	21.4647	0.7349
4.200	29.2475	21.4834	0.7345
4.205	29.2445	21.4962	0.7351
4.210	29.1214	21.5247	0.7391
4.215	29.1973	21.5300	0.7374
4.220	29.1541	21.5507	0.7392
4.225	29.2446	21.5398	0.7365

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.230	29.1481	21.5468	0.7392
4.235	29.0892	21.5582	0.7411
4.240	29.1010	21.6581	0.7442
4.245	29.1329	21.5949	0.7413
4.250	29.0741	21.5870	0.7425
4.255	29.0970	21.6296	0.7434
4.260	29.0112	21.6279	0.7455
4.265	29.0759	21.6163	0.7434
4.270	28.9927	21.6319	0.7461
4.275	28.9645	21.6490	0.7474
4.280	28.9580	21.6757	0.7485
4.285	28.9313	21.6990	0.7500
4.290	28.8923	21.6896	0.7507
4.295	28.8823	21.6860	0.7508
4.300	28.8788	21.7307	0.7525
4.305	28.8423	21.7331	0.7535
4.310	28.8324	21.7172	0.7532
4.315	28.8153	21.7327	0.7542
4.320	28.8178	21.7279	0.7540
4.325	28.7883	21.7435	0.7553
4.330	28.7994	21.7398	0.7549
4.335	28.7712	21.7513	0.7560
4.340	28.7643	21.7187	0.7551
4.345	28.7202	21.7247	0.7564
4.350	28.7128	21.7272	0.7567
4.355	28.6827	21.7472	0.7582
4.360	28.6464	21.7181	0.7581
4.365	28.5864	21.7380	0.7604
4.370	28.6176	21.7176	0.7589
4.375	28.5354	21.7669	0.7628
4.380	28.5398	21.7484	0.7620
4.385	28.4241	21.7326	0.7646
4.390	28.4581	21.7021	0.7626
4.395	28.3945	21.7390	0.7656
4.400	28.3676	21.7688	0.7674
4.405	28.3009	21.7694	0.7692
4.410	28.3217	21.7921	0.7694
4.415	28.2864	21.7614	0.7693
4.420	28.2487	21.8377	0.7731
4.425	28.2814	21.8271	0.7718
4.430	28.2087	21.7907	0.7725
4.435	28.1361	21.7634	0.7735
4.440	28.1664	21.8015	0.7740
4.445	28.0564	21.7810	0.7763
4.450	28.0855	21.7240	0.7735
4.455	27.9863	21.7974	0.7789

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.460	28.0141	21.6812	0.7739
4.465	28.0063	21.6097	0.7716
4.470	28.0315	21.6501	0.7723
4.475	28.0433	21.5981	0.7702
4.480	28.0749	21.6160	0.7699
4.485	27.9338	21.5698	0.7722
4.490	27.9633	21.6100	0.7728
4.495	27.9702	21.6082	0.7725
4.500	27.9240	21.6154	0.7741
4.505	27.8623	21.6238	0.7761
4.510	27.7585	21.6475	0.7799
4.515	27.7935	21.7262	0.7817
4.520	27.7645	21.7172	0.7822
4.525	27.7871	21.7160	0.7815
4.530	27.6742	21.6936	0.7839
4.535	27.6802	21.7264	0.7849
4.540	27.6708	21.6903	0.7839
4.545	27.6050	21.7018	0.7862
4.550	27.6001	21.6500	0.7844
4.555	27.5589	21.6981	0.7873
4.560	27.5590	21.7047	0.7876
4.565	27.5293	21.7360	0.7896
4.570	27.4999	21.6985	0.7890
4.575	27.4908	21.7043	0.7895
4.580	27.4861	21.7292	0.7906
4.585	27.4518	21.6988	0.7904
4.590	27.4171	21.6900	0.7911
4.595	27.3964	21.7259	0.7930
4.600	27.3666	21.7101	0.7933
4.605	27.3331	21.7030	0.7940
4.610	27.3216	21.6990	0.7942
4.615	27.2942	21.6784	0.7942
4.620	27.2892	21.6627	0.7938
4.625	27.2675	21.6986	0.7958
4.630	27.2397	21.6650	0.7953
4.635	27.1931	21.6483	0.7961
4.640	27.1492	21.6684	0.7981
4.645	27.1529	21.6621	0.7978
4.650	27.1165	21.6545	0.7986
4.655	27.0758	21.5984	0.7977
4.660	27.1294	21.6575	0.7983
4.665	27.0929	21.6345	0.7985
4.670	26.9521	21.6548	0.8035
4.675	26.9498	21.5721	0.8005
4.680	26.9565	21.5985	0.8012
4.685	26.9515	21.5940	0.8012

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.690	26.9483	21.6573	0.8037
4.695	26.9253	21.5543	0.8005
4.700	26.8815	21.6448	0.8052
4.705	26.7953	21.4863	0.8019
4.710	26.6873	21.6553	0.8114
4.715	26.7585	21.6397	0.8087
4.720	26.7298	21.5815	0.8074
4.725	26.7377	21.6784	0.8108
4.730	26.7472	21.6362	0.8089
4.735	26.7525	21.6830	0.8105
4.740	26.7578	21.6497	0.8091
4.745	26.6633	21.6820	0.8132
4.750	26.7364	21.6325	0.8091
4.755	26.7266	21.6975	0.8118
4.760	26.6697	21.6463	0.8116
4.765	26.7519	21.6538	0.8094
4.770	26.7190	21.6909	0.8118
4.775	26.5454	21.5929	0.8134
4.780	26.5944	21.6008	0.8122
4.785	26.6209	21.8320	0.8201
4.790	26.5512	21.5612	0.8121
4.795	26.4726	21.6059	0.8162
4.800	26.4814	21.6244	0.8166
4.805	26.4969	21.5346	0.8127
4.810	26.4413	21.6665	0.8194
4.815	26.4705	21.5312	0.8134
4.820	26.3446	21.6724	0.8227
4.825	26.4021	21.6577	0.8203
4.830	26.3095	21.6937	0.8246
4.835	26.3188	21.7119	0.8250
4.840	26.3324	21.6947	0.8239
4.845	26.3453	21.6601	0.8222
4.850	26.2554	21.7168	0.8271
4.855	26.3367	21.7300	0.8251
4.860	26.2897	21.7002	0.8254
4.865	26.3291	21.6775	0.8233
4.870	26.2530	21.6997	0.8266
4.875	26.2654	21.7217	0.8270
4.880	26.2609	21.7523	0.8283
4.885	26.2190	21.6927	0.8274
4.890	26.2948	21.7421	0.8269
4.895	26.1872	21.7148	0.8292
4.900	26.2085	21.7393	0.8295
4.905	26.1902	21.7270	0.8296
4.910	26.1329	21.7267	0.8314
4.915	26.1247	21.7319	0.8319

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.920	26.1125	21.7171	0.8317
4.925	26.1021	21.7193	0.8321
4.930	26.0478	21.7086	0.8334
4.935	26.0470	21.7126	0.8336
4.940	26.0347	21.7163	0.8341
4.945	26.0068	21.7332	0.8357
4.950	26.0087	21.7236	0.8352
4.955	25.9973	21.7465	0.8365
4.960	25.9537	21.7262	0.8371
4.965	25.9458	21.7426	0.8380
4.970	25.9301	21.7681	0.8395
4.975	25.9401	21.7594	0.8388
4.980	25.8590	21.7806	0.8423
4.985	25.8297	21.7629	0.8426
4.990	25.8230	21.7962	0.8441
4.995	25.7249	21.8162	0.8481
5.000	25.7138	21.7972	0.8477
5.005	25.6662	21.8172	0.8500
5.010	25.6122	21.7922	0.8509
5.015	25.5815	21.8415	0.8538
5.020	25.5279	21.7827	0.8533
5.025	25.4183	21.7898	0.8572
5.030	25.3489	21.7558	0.8583
5.035	25.3660	21.7645	0.8580
5.040	25.2973	21.7170	0.8585
5.045	25.2992	21.6540	0.8559
5.050	25.2466	21.5690	0.8543
5.055	25.2979	21.5907	0.8535
5.060	25.2513	21.5594	0.8538
5.065	25.2689	21.5135	0.8514
5.070	25.2573	21.5719	0.8541
5.075	25.2520	21.5331	0.8527
5.080	25.2038	21.5145	0.8536
5.085	25.1945	21.4975	0.8533
5.090	25.2149	21.5289	0.8538
5.095	25.2384	21.5618	0.8543
5.100	25.2327	21.5615	0.8545
5.105	25.2131	21.6464	0.8585
5.110	25.1667	21.5475	0.8562
5.115	25.1257	21.5512	0.8577
5.120	25.1210	21.5561	0.8581
5.125	25.0974	21.5730	0.8596
5.130	25.0843	21.5463	0.8590
5.135	25.0173	21.5478	0.8613
5.140	25.0178	21.5822	0.8627
5.145	24.9666	21.4934	0.8609

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.150	24.9653	21.5708	0.8640
5.155	24.9332	21.5289	0.8635
5.160	24.9114	21.5158	0.8637
5.165	24.8657	21.5196	0.8654
5.170	24.8587	21.4879	0.8644
5.175	24.8120	21.4825	0.8658
5.180	24.8147	21.4795	0.8656
5.185	24.7829	21.4759	0.8666
5.190	24.7883	21.4568	0.8656
5.195	24.7757	21.4550	0.8660
5.200	24.7726	21.4535	0.8660
5.205	24.7664	21.4557	0.8663
5.210	24.7508	21.4521	0.8667
5.215	24.7444	21.4602	0.8673
5.220	24.7107	21.4531	0.8682
5.225	24.6744	21.4524	0.8694
5.230	24.6919	21.4405	0.8683
5.235	24.6567	21.4717	0.8708
5.240	24.6204	21.4508	0.8713
5.245	24.6318	21.4627	0.8713
5.250	24.5946	21.4858	0.8736
5.255	24.5371	21.4911	0.8759
5.260	24.6112	21.4570	0.8718
5.265	24.6125	21.5142	0.8741
5.270	24.5316	21.4710	0.8752
5.275	24.4934	21.4340	0.8751
5.280	24.5097	21.4698	0.8760
5.285	24.5181	21.4373	0.8743
5.290	24.4620	21.4184	0.8756
5.295	24.4956	21.3537	0.8717
5.300	24.4387	21.4145	0.8763
5.305	24.4768	21.4158	0.8749
5.310	24.3344	21.4104	0.8798
5.315	24.3834	21.4156	0.8783
5.320	24.3261	21.3673	0.8784
5.325	24.3487	21.3996	0.8789
5.330	24.3135	21.3941	0.8799
5.335	24.2306	21.4093	0.8836
5.340	24.2239	21.4616	0.8860
5.345	24.1983	21.4839	0.8878
5.350	24.2642	21.3919	0.8816
5.355	24.1667	21.4347	0.8870
5.360	24.1359	21.3321	0.8838
5.365	24.2485	21.4013	0.8826
5.370	24.0917	21.3462	0.8860
5.375	24.1412	21.4120	0.8869

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.380	24.1495	21.4140	0.8867
5.385	24.0711	21.3680	0.8877
5.390	24.0656	21.4481	0.8912
5.395	24.1159	21.4278	0.8885
5.400	24.0377	21.4192	0.8911
5.405	24.0249	21.4007	0.8908
5.410	24.0014	21.4240	0.8926
5.415	24.0050	21.3538	0.8896
5.420	23.9498	21.4364	0.8951
5.425	23.9524	21.4635	0.8961
5.430	23.9582	21.3950	0.8930
5.435	23.9381	21.3948	0.8938
5.440	23.8601	21.4494	0.8990
5.445	23.9278	21.4232	0.8953
5.450	23.8274	21.4203	0.8990
5.455	23.8738	21.4447	0.8983
5.460	23.8383	21.4186	0.8985
5.465	23.8589	21.4862	0.9006
5.470	23.7456	21.4503	0.9033
5.475	23.7794	21.4256	0.9010
5.480	23.7477	21.4720	0.9042
5.485	23.7537	21.4332	0.9023
5.490	23.6865	21.5095	0.9081
5.495	23.7451	21.4417	0.9030
5.500	23.7160	21.4860	0.9060
5.505	23.7182	21.4977	0.9064
5.510	23.7136	21.4743	0.9056
5.515	23.7082	21.4547	0.9049
5.520	23.6664	21.4513	0.9064
5.525	23.6899	21.4471	0.9053
5.530	23.6660	21.4536	0.9065
5.535	23.6664	21.4303	0.9055
5.540	23.6329	21.4616	0.9081
5.545	23.5840	21.4315	0.9087
5.550	23.5956	21.4383	0.9086
5.555	23.6010	21.4211	0.9076
5.560	23.5798	21.4299	0.9088
5.565	23.5897	21.4247	0.9082
5.570	23.5817	21.4322	0.9089
5.575	23.5853	21.4252	0.9084
5.580	23.5801	21.4553	0.9099
5.585	23.5609	21.4421	0.9101
5.590	23.5754	21.4193	0.9085
5.595	23.5539	21.4167	0.9093
5.600	23.5677	21.4204	0.9089
5.605	23.5783	21.3820	0.9068

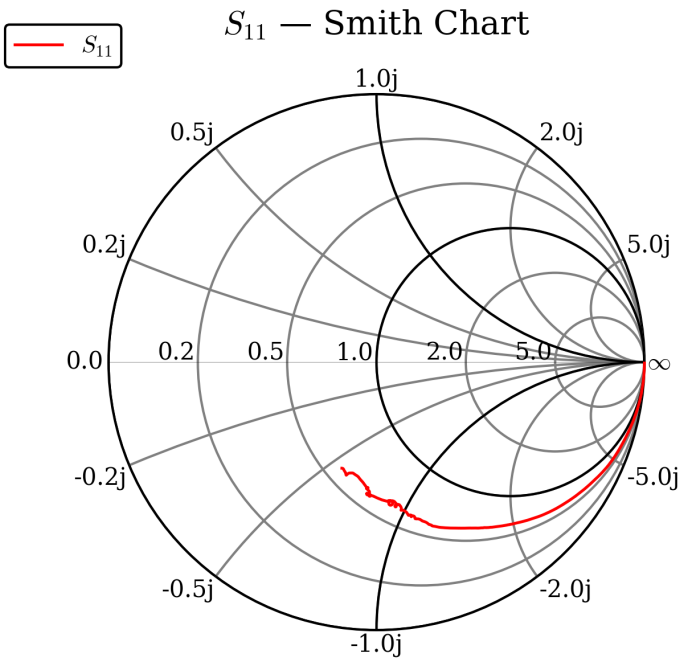
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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.610	23.5252	21.3517	0.9076
5.615	23.5057	21.3133	0.9067
5.620	23.4869	21.3197	0.9077
5.625	23.4744	21.3528	0.9096
5.630	23.4330	21.3047	0.9092
5.635	23.4347	21.2767	0.9079
5.640	23.4118	21.2332	0.9069
5.645	23.3534	21.2327	0.9092
5.650	23.3770	21.2790	0.9103
5.655	23.3289	21.2549	0.9111
5.660	23.2977	21.2718	0.9130
5.665	23.3730	21.2383	0.9087
5.670	23.2095	21.2811	0.9169
5.675	23.2466	21.2004	0.9120
5.680	23.1967	21.2156	0.9146
5.685	23.2502	21.1728	0.9107
5.690	23.1950	21.1735	0.9128
5.695	23.1750	21.2406	0.9165
5.700	23.1340	21.1408	0.9138
5.705	23.0455	21.1558	0.9180
5.710	23.1204	21.1255	0.9137
5.715	23.0543	21.0877	0.9147
5.720	23.0332	21.1220	0.9170
5.725	22.9845	21.1399	0.9197
5.730	22.9748	21.1138	0.9190
5.735	22.9587	21.1080	0.9194
5.740	22.9442	21.1256	0.9207
5.745	22.9452	21.1394	0.9213
5.750	22.8725	21.1083	0.9229
5.755	22.8937	21.1078	0.9220
5.760	22.8119	21.0750	0.9239
5.765	22.8631	21.0601	0.9211
5.770	22.8296	21.0380	0.9215
5.775	22.8195	21.0567	0.9228
5.780	22.7626	21.0889	0.9265
5.785	22.7221	21.0577	0.9268
5.790	22.6943	21.1014	0.9298
5.795	22.7033	21.1107	0.9299
5.800	22.6586	21.0781	0.9302
5.805	22.6427	21.0605	0.9301
5.810	22.6384	21.0986	0.9320
5.815	22.6246	21.0922	0.9323
5.820	22.6055	21.0662	0.9319
5.825	22.5860	21.0748	0.9331
5.830	22.5403	21.0792	0.9352
5.835	22.5536	21.0603	0.9338

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.840	22.5263	21.0689	0.9353
5.845	22.4937	21.0740	0.9369
5.850	22.4847	21.0499	0.9362
5.855	22.4656	21.0560	0.9373
5.860	22.4226	21.0625	0.9393
5.865	22.4408	21.0639	0.9386
5.870	22.4072	21.0376	0.9389
5.875	22.3917	21.0406	0.9397
5.880	22.3756	21.0412	0.9404
5.885	22.3813	21.0003	0.9383
5.890	22.2926	21.0674	0.9450
5.895	22.3053	21.0268	0.9427
5.900	22.2954	20.9776	0.9409
5.905	22.2884	21.0612	0.9449
5.910	22.2331	20.9625	0.9429
5.915	22.2247	20.9969	0.9448
5.920	22.2140	20.9743	0.9442
5.925	22.1946	20.9626	0.9445
5.930	22.1224	21.0023	0.9494
5.935	22.1093	20.9642	0.9482
5.940	22.0960	20.9862	0.9498
5.945	22.1671	20.9527	0.9452
5.950	22.0817	20.9992	0.9510
5.955	22.0806	21.0063	0.9513
5.960	22.0483	20.9951	0.9522
5.965	22.0310	20.9822	0.9524
5.970	22.0251	21.0254	0.9546
5.975	22.0524	20.8931	0.9474
5.980	22.1521	20.9580	0.9461
5.985	21.9597	20.9928	0.9560
5.990	21.9389	20.9294	0.9540
5.995	21.9404	20.9081	0.9530
6.000	21.9869	20.8874	0.9500

1.2 S_{11} — Smith Chart



Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
0.005	0.9999	-0.0034	-0.00
0.010	0.9999	-0.0065	-0.00
0.015	1.0000	-0.0097	0.00
0.020	0.9998	-0.0131	-0.00
0.025	0.9997	-0.0161	-0.00
0.030	0.9996	-0.0196	-0.00
0.035	0.9995	-0.0227	-0.00
0.040	0.9993	-0.0261	-0.00
0.045	0.9991	-0.0291	-0.00
0.050	0.9990	-0.0325	-0.00
0.055	0.9989	-0.0357	-0.00
0.060	0.9986	-0.0390	-0.01
0.065	0.9984	-0.0423	-0.01
0.070	0.9981	-0.0454	-0.01
0.075	0.9980	-0.0486	-0.01
0.080	0.9975	-0.0519	-0.01
0.085	0.9974	-0.0552	-0.01
0.090	0.9970	-0.0584	-0.01
0.095	0.9966	-0.0616	-0.01
0.100	0.9963	-0.0648	-0.01
0.105	0.9960	-0.0679	-0.01
0.110	0.9956	-0.0712	-0.02
0.115	0.9952	-0.0746	-0.02

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.120	0.9950	-0.0776	-0.02
0.125	0.9944	-0.0809	-0.02
0.130	0.9938	-0.0839	-0.02
0.135	0.9935	-0.0872	-0.02
0.140	0.9930	-0.0904	-0.03
0.145	0.9925	-0.0936	-0.03
0.150	0.9920	-0.0968	-0.03
0.155	0.9915	-0.1000	-0.03
0.160	0.9910	-0.1032	-0.03
0.165	0.9905	-0.1061	-0.03
0.170	0.9899	-0.1094	-0.04
0.175	0.9893	-0.1127	-0.04
0.180	0.9887	-0.1157	-0.04
0.185	0.9880	-0.1189	-0.04
0.190	0.9873	-0.1221	-0.05
0.195	0.9866	-0.1251	-0.05
0.200	0.9859	-0.1283	-0.05
0.205	0.9853	-0.1315	-0.05
0.210	0.9846	-0.1345	-0.05
0.215	0.9838	-0.1378	-0.06
0.220	0.9829	-0.1408	-0.06
0.225	0.9824	-0.1440	-0.06
0.230	0.9815	-0.1471	-0.07
0.235	0.9807	-0.1502	-0.07
0.240	0.9798	-0.1533	-0.07
0.245	0.9791	-0.1565	-0.07
0.250	0.9782	-0.1595	-0.08
0.255	0.9773	-0.1626	-0.08
0.260	0.9764	-0.1657	-0.08
0.265	0.9755	-0.1688	-0.09
0.270	0.9746	-0.1719	-0.09
0.275	0.9736	-0.1749	-0.09
0.280	0.9727	-0.1780	-0.10
0.285	0.9718	-0.1810	-0.10
0.290	0.9709	-0.1840	-0.10
0.295	0.9698	-0.1871	-0.11
0.300	0.9689	-0.1900	-0.11
0.305	0.9678	-0.1931	-0.11
0.310	0.9667	-0.1961	-0.12
0.315	0.9656	-0.1992	-0.12
0.320	0.9647	-0.2021	-0.13
0.325	0.9636	-0.2051	-0.13
0.330	0.9624	-0.2081	-0.13
0.335	0.9613	-0.2111	-0.14
0.340	0.9602	-0.2139	-0.14
0.345	0.9591	-0.2169	-0.15

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.350	0.9579	-0.2198	-0.15
0.355	0.9569	-0.2226	-0.15
0.360	0.9556	-0.2256	-0.16
0.365	0.9545	-0.2284	-0.16
0.370	0.9531	-0.2314	-0.17
0.375	0.9519	-0.2343	-0.17
0.380	0.9506	-0.2372	-0.18
0.385	0.9494	-0.2401	-0.18
0.390	0.9482	-0.2429	-0.19
0.395	0.9471	-0.2457	-0.19
0.400	0.9458	-0.2486	-0.19
0.405	0.9445	-0.2513	-0.20
0.410	0.9432	-0.2541	-0.20
0.415	0.9420	-0.2569	-0.21
0.420	0.9407	-0.2597	-0.21
0.425	0.9394	-0.2623	-0.22
0.430	0.9381	-0.2650	-0.22
0.435	0.9368	-0.2677	-0.23
0.440	0.9355	-0.2705	-0.23
0.445	0.9342	-0.2730	-0.24
0.450	0.9328	-0.2758	-0.24
0.455	0.9315	-0.2785	-0.24
0.460	0.9301	-0.2811	-0.25
0.465	0.9288	-0.2838	-0.25
0.470	0.9274	-0.2863	-0.26
0.475	0.9259	-0.2891	-0.26
0.480	0.9245	-0.2915	-0.27
0.485	0.9233	-0.2941	-0.27
0.490	0.9219	-0.2967	-0.28
0.495	0.9204	-0.2994	-0.28
0.500	0.9190	-0.3018	-0.29
0.505	0.9177	-0.3045	-0.29
0.510	0.9163	-0.3069	-0.30
0.515	0.9149	-0.3094	-0.30
0.520	0.9135	-0.3119	-0.31
0.525	0.9121	-0.3144	-0.31
0.530	0.9108	-0.3168	-0.32
0.535	0.9094	-0.3192	-0.32
0.540	0.9080	-0.3218	-0.32
0.545	0.9066	-0.3242	-0.33
0.550	0.9051	-0.3267	-0.33
0.555	0.9036	-0.3291	-0.34
0.560	0.9021	-0.3315	-0.34
0.565	0.9006	-0.3340	-0.35
0.570	0.8991	-0.3362	-0.36
0.575	0.8977	-0.3388	-0.36

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.580	0.8962	-0.3410	-0.36
0.585	0.8948	-0.3435	-0.37
0.590	0.8932	-0.3459	-0.37
0.595	0.8917	-0.3481	-0.38
0.600	0.8901	-0.3504	-0.39
0.605	0.8886	-0.3527	-0.39
0.610	0.8871	-0.3551	-0.40
0.615	0.8855	-0.3573	-0.40
0.620	0.8841	-0.3596	-0.41
0.625	0.8824	-0.3619	-0.41
0.630	0.8809	-0.3641	-0.42
0.635	0.8794	-0.3664	-0.42
0.640	0.8777	-0.3687	-0.43
0.645	0.8761	-0.3709	-0.43
0.650	0.8746	-0.3732	-0.44
0.655	0.8729	-0.3753	-0.44
0.660	0.8713	-0.3774	-0.45
0.665	0.8697	-0.3797	-0.46
0.670	0.8679	-0.3819	-0.46
0.675	0.8663	-0.3839	-0.47
0.680	0.8647	-0.3860	-0.47
0.685	0.8630	-0.3881	-0.48
0.690	0.8613	-0.3901	-0.49
0.695	0.8597	-0.3921	-0.49
0.700	0.8579	-0.3941	-0.50
0.705	0.8560	-0.3963	-0.51
0.710	0.8544	-0.3984	-0.51
0.715	0.8526	-0.4003	-0.52
0.720	0.8509	-0.4023	-0.53
0.725	0.8491	-0.4043	-0.53
0.730	0.8473	-0.4064	-0.54
0.735	0.8455	-0.4084	-0.55
0.740	0.8436	-0.4101	-0.56
0.745	0.8418	-0.4121	-0.56
0.750	0.8401	-0.4140	-0.57
0.755	0.8384	-0.4157	-0.58
0.760	0.8365	-0.4176	-0.58
0.765	0.8346	-0.4195	-0.59
0.770	0.8327	-0.4213	-0.60
0.775	0.8308	-0.4233	-0.61
0.780	0.8291	-0.4251	-0.61
0.785	0.8271	-0.4270	-0.62
0.790	0.8254	-0.4288	-0.63
0.795	0.8234	-0.4304	-0.64
0.800	0.8216	-0.4322	-0.65
0.805	0.8197	-0.4340	-0.65

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.810	0.8178	-0.4358	-0.66
0.815	0.8160	-0.4374	-0.67
0.820	0.8142	-0.4391	-0.68
0.825	0.8125	-0.4410	-0.68
0.830	0.8106	-0.4427	-0.69
0.835	0.8086	-0.4445	-0.70
0.840	0.8069	-0.4463	-0.70
0.845	0.8050	-0.4480	-0.71
0.850	0.8031	-0.4497	-0.72
0.855	0.8013	-0.4514	-0.73
0.860	0.7993	-0.4532	-0.73
0.865	0.7975	-0.4549	-0.74
0.870	0.7955	-0.4569	-0.75
0.875	0.7935	-0.4585	-0.76
0.880	0.7917	-0.4603	-0.76
0.885	0.7897	-0.4620	-0.77
0.890	0.7877	-0.4638	-0.78
0.895	0.7858	-0.4655	-0.79
0.900	0.7838	-0.4671	-0.80
0.905	0.7818	-0.4689	-0.80
0.910	0.7799	-0.4706	-0.81
0.915	0.7779	-0.4724	-0.82
0.920	0.7758	-0.4740	-0.83
0.925	0.7737	-0.4757	-0.84
0.930	0.7717	-0.4774	-0.84
0.935	0.7697	-0.4791	-0.85
0.940	0.7676	-0.4808	-0.86
0.945	0.7654	-0.4824	-0.87
0.950	0.7635	-0.4842	-0.88
0.955	0.7614	-0.4857	-0.89
0.960	0.7593	-0.4874	-0.89
0.965	0.7572	-0.4890	-0.90
0.970	0.7552	-0.4907	-0.91
0.975	0.7531	-0.4923	-0.92
0.980	0.7511	-0.4938	-0.93
0.985	0.7489	-0.4954	-0.93
0.990	0.7468	-0.4969	-0.94
0.995	0.7447	-0.4986	-0.95
1.000	0.7426	-0.5001	-0.96
1.005	0.7404	-0.5017	-0.97
1.010	0.7384	-0.5030	-0.98
1.015	0.7363	-0.5045	-0.99
1.020	0.7341	-0.5062	-1.00
1.025	0.7320	-0.5076	-1.00
1.030	0.7298	-0.5091	-1.01
1.035	0.7278	-0.5105	-1.02

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.040	0.7257	-0.5120	-1.03
1.045	0.7236	-0.5135	-1.04
1.050	0.7215	-0.5148	-1.05
1.055	0.7196	-0.5161	-1.06
1.060	0.7175	-0.5175	-1.06
1.065	0.7154	-0.5189	-1.07
1.070	0.7133	-0.5204	-1.08
1.075	0.7112	-0.5218	-1.09
1.080	0.7092	-0.5231	-1.10
1.085	0.7071	-0.5243	-1.11
1.090	0.7051	-0.5257	-1.12
1.095	0.7030	-0.5270	-1.12
1.100	0.7009	-0.5282	-1.13
1.105	0.6990	-0.5296	-1.14
1.110	0.6969	-0.5307	-1.15
1.115	0.6949	-0.5320	-1.16
1.120	0.6929	-0.5332	-1.17
1.125	0.6909	-0.5343	-1.18
1.130	0.6889	-0.5355	-1.18
1.135	0.6870	-0.5367	-1.19
1.140	0.6851	-0.5379	-1.20
1.145	0.6831	-0.5391	-1.21
1.150	0.6811	-0.5402	-1.22
1.155	0.6791	-0.5413	-1.23
1.160	0.6772	-0.5423	-1.23
1.165	0.6753	-0.5435	-1.24
1.170	0.6736	-0.5447	-1.25
1.175	0.6717	-0.5456	-1.26
1.180	0.6697	-0.5467	-1.26
1.185	0.6680	-0.5477	-1.27
1.190	0.6662	-0.5486	-1.28
1.195	0.6644	-0.5496	-1.29
1.200	0.6626	-0.5505	-1.30
1.205	0.6608	-0.5514	-1.30
1.210	0.6590	-0.5523	-1.31
1.215	0.6571	-0.5532	-1.32
1.220	0.6554	-0.5542	-1.33
1.225	0.6534	-0.5551	-1.34
1.230	0.6516	-0.5559	-1.35
1.235	0.6499	-0.5568	-1.35
1.240	0.6481	-0.5577	-1.36
1.245	0.6464	-0.5586	-1.37
1.250	0.6446	-0.5594	-1.38
1.255	0.6428	-0.5602	-1.38
1.260	0.6412	-0.5610	-1.39
1.265	0.6394	-0.5618	-1.40

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.270	0.6377	-0.5626	-1.41
1.275	0.6359	-0.5633	-1.42
1.280	0.6342	-0.5643	-1.42
1.285	0.6324	-0.5652	-1.43
1.290	0.6306	-0.5659	-1.44
1.295	0.6288	-0.5667	-1.45
1.300	0.6271	-0.5674	-1.46
1.305	0.6254	-0.5682	-1.46
1.310	0.6237	-0.5688	-1.47
1.315	0.6219	-0.5696	-1.48
1.320	0.6203	-0.5704	-1.49
1.325	0.6187	-0.5709	-1.50
1.330	0.6169	-0.5716	-1.50
1.335	0.6154	-0.5722	-1.51
1.340	0.6136	-0.5730	-1.52
1.345	0.6119	-0.5737	-1.53
1.350	0.6102	-0.5742	-1.54
1.355	0.6085	-0.5750	-1.54
1.360	0.6067	-0.5756	-1.55
1.365	0.6049	-0.5763	-1.56
1.370	0.6032	-0.5768	-1.57
1.375	0.6015	-0.5775	-1.58
1.380	0.5997	-0.5781	-1.59
1.385	0.5979	-0.5787	-1.60
1.390	0.5962	-0.5794	-1.60
1.395	0.5943	-0.5798	-1.62
1.400	0.5924	-0.5804	-1.62
1.405	0.5906	-0.5812	-1.63
1.410	0.5886	-0.5818	-1.64
1.415	0.5867	-0.5824	-1.65
1.420	0.5848	-0.5830	-1.66
1.425	0.5830	-0.5837	-1.67
1.430	0.5811	-0.5843	-1.68
1.435	0.5790	-0.5850	-1.69
1.440	0.5771	-0.5857	-1.70
1.445	0.5751	-0.5864	-1.71
1.450	0.5731	-0.5870	-1.72
1.455	0.5713	-0.5877	-1.73
1.460	0.5692	-0.5882	-1.74
1.465	0.5672	-0.5889	-1.75
1.470	0.5652	-0.5894	-1.76
1.475	0.5632	-0.5900	-1.77
1.480	0.5611	-0.5906	-1.78
1.485	0.5590	-0.5912	-1.79
1.490	0.5568	-0.5919	-1.80
1.495	0.5548	-0.5925	-1.81

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.500	0.5527	-0.5931	-1.82
1.505	0.5505	-0.5936	-1.83
1.510	0.5483	-0.5941	-1.85
1.515	0.5462	-0.5947	-1.86
1.520	0.5440	-0.5952	-1.87
1.525	0.5418	-0.5960	-1.88
1.530	0.5396	-0.5963	-1.89
1.535	0.5374	-0.5970	-1.90
1.540	0.5352	-0.5975	-1.92
1.545	0.5330	-0.5981	-1.93
1.550	0.5308	-0.5985	-1.94
1.555	0.5285	-0.5990	-1.95
1.560	0.5261	-0.5995	-1.96
1.565	0.5237	-0.6000	-1.98
1.570	0.5215	-0.6006	-1.99
1.575	0.5192	-0.6011	-2.00
1.580	0.5169	-0.6017	-2.01
1.585	0.5146	-0.6021	-2.02
1.590	0.5124	-0.6027	-2.04
1.595	0.5102	-0.6033	-2.05
1.600	0.5080	-0.6039	-2.06
1.605	0.5057	-0.6043	-2.07
1.610	0.5035	-0.6048	-2.08
1.615	0.5012	-0.6053	-2.09
1.620	0.4989	-0.6055	-2.11
1.625	0.4966	-0.6061	-2.12
1.630	0.4945	-0.6067	-2.13
1.635	0.4920	-0.6069	-2.14
1.640	0.4897	-0.6074	-2.16
1.645	0.4874	-0.6078	-2.17
1.650	0.4851	-0.6084	-2.18
1.655	0.4829	-0.6087	-2.19
1.660	0.4805	-0.6092	-2.20
1.665	0.4784	-0.6097	-2.21
1.670	0.4761	-0.6098	-2.23
1.675	0.4738	-0.6103	-2.24
1.680	0.4717	-0.6107	-2.25
1.685	0.4695	-0.6111	-2.26
1.690	0.4673	-0.6114	-2.28
1.695	0.4653	-0.6117	-2.29
1.700	0.4630	-0.6119	-2.30
1.705	0.4610	-0.6121	-2.31
1.710	0.4586	-0.6126	-2.32
1.715	0.4565	-0.6128	-2.34
1.720	0.4541	-0.6132	-2.35
1.725	0.4520	-0.6135	-2.36

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.730	0.4500	-0.6137	-2.37
1.735	0.4477	-0.6139	-2.39
1.740	0.4458	-0.6142	-2.40
1.745	0.4437	-0.6146	-2.41
1.750	0.4419	-0.6148	-2.42
1.755	0.4398	-0.6151	-2.43
1.760	0.4380	-0.6153	-2.44
1.765	0.4361	-0.6157	-2.45
1.770	0.4344	-0.6159	-2.46
1.775	0.4327	-0.6159	-2.47
1.780	0.4310	-0.6161	-2.48
1.785	0.4294	-0.6161	-2.49
1.790	0.4274	-0.6162	-2.50
1.795	0.4256	-0.6163	-2.51
1.800	0.4237	-0.6165	-2.52
1.805	0.4220	-0.6168	-2.53
1.810	0.4202	-0.6169	-2.54
1.815	0.4186	-0.6172	-2.55
1.820	0.4170	-0.6173	-2.56
1.825	0.4155	-0.6174	-2.57
1.830	0.4140	-0.6174	-2.58
1.835	0.4125	-0.6176	-2.58
1.840	0.4108	-0.6176	-2.60
1.845	0.4095	-0.6176	-2.60
1.850	0.4079	-0.6175	-2.61
1.855	0.4064	-0.6174	-2.63
1.860	0.4049	-0.6174	-2.64
1.865	0.4034	-0.6174	-2.64
1.870	0.4018	-0.6176	-2.65
1.875	0.4002	-0.6177	-2.66
1.880	0.3990	-0.6181	-2.67
1.885	0.3978	-0.6183	-2.67
1.890	0.3967	-0.6183	-2.68
1.895	0.3953	-0.6183	-2.69
1.900	0.3939	-0.6183	-2.70
1.905	0.3928	-0.6182	-2.70
1.910	0.3915	-0.6184	-2.71
1.915	0.3901	-0.6185	-2.72
1.920	0.3889	-0.6187	-2.72
1.925	0.3878	-0.6187	-2.73
1.930	0.3865	-0.6189	-2.74
1.935	0.3856	-0.6190	-2.74
1.940	0.3844	-0.6190	-2.75
1.945	0.3832	-0.6190	-2.76
1.950	0.3821	-0.6191	-2.76
1.955	0.3810	-0.6192	-2.77

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.960	0.3799	-0.6191	-2.78
1.965	0.3788	-0.6193	-2.78
1.970	0.3776	-0.6193	-2.79
1.975	0.3767	-0.6192	-2.80
1.980	0.3756	-0.6192	-2.80
1.985	0.3746	-0.6191	-2.81
1.990	0.3736	-0.6191	-2.82
1.995	0.3724	-0.6191	-2.82
2.000	0.3713	-0.6191	-2.83
2.005	0.3701	-0.6191	-2.84
2.010	0.3689	-0.6191	-2.84
2.015	0.3680	-0.6193	-2.85
2.020	0.3669	-0.6192	-2.86
2.025	0.3657	-0.6192	-2.86
2.030	0.3645	-0.6193	-2.87
2.035	0.3634	-0.6194	-2.88
2.040	0.3622	-0.6192	-2.88
2.045	0.3611	-0.6192	-2.89
2.050	0.3599	-0.6193	-2.90
2.055	0.3588	-0.6193	-2.91
2.060	0.3575	-0.6192	-2.91
2.065	0.3562	-0.6193	-2.92
2.070	0.3549	-0.6193	-2.93
2.075	0.3537	-0.6193	-2.94
2.080	0.3524	-0.6193	-2.94
2.085	0.3511	-0.6191	-2.95
2.090	0.3496	-0.6192	-2.96
2.095	0.3482	-0.6191	-2.97
2.100	0.3467	-0.6191	-2.98
2.105	0.3453	-0.6192	-2.99
2.110	0.3438	-0.6192	-3.00
2.115	0.3423	-0.6192	-3.01
2.120	0.3408	-0.6194	-3.01
2.125	0.3392	-0.6193	-3.02
2.130	0.3376	-0.6194	-3.03
2.135	0.3360	-0.6195	-3.04
2.140	0.3344	-0.6195	-3.05
2.145	0.3327	-0.6195	-3.06
2.150	0.3312	-0.6196	-3.07
2.155	0.3295	-0.6195	-3.08
2.160	0.3277	-0.6194	-3.09
2.165	0.3261	-0.6195	-3.10
2.170	0.3244	-0.6196	-3.11
2.175	0.3226	-0.6195	-3.12
2.180	0.3209	-0.6195	-3.13
2.185	0.3191	-0.6195	-3.14

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.190	0.3173	-0.6195	-3.15
2.195	0.3154	-0.6195	-3.16
2.200	0.3136	-0.6196	-3.17
2.205	0.3118	-0.6195	-3.18
2.210	0.3099	-0.6195	-3.19
2.215	0.3080	-0.6194	-3.20
2.220	0.3062	-0.6195	-3.21
2.225	0.3044	-0.6194	-3.22
2.230	0.3024	-0.6194	-3.23
2.235	0.3005	-0.6194	-3.24
2.240	0.2986	-0.6194	-3.25
2.245	0.2968	-0.6193	-3.26
2.250	0.2949	-0.6195	-3.27
2.255	0.2930	-0.6194	-3.28
2.260	0.2913	-0.6195	-3.29
2.265	0.2894	-0.6194	-3.30
2.270	0.2875	-0.6194	-3.31
2.275	0.2857	-0.6193	-3.32
2.280	0.2837	-0.6193	-3.33
2.285	0.2819	-0.6193	-3.34
2.290	0.2801	-0.6191	-3.36
2.295	0.2783	-0.6192	-3.36
2.300	0.2765	-0.6192	-3.37
2.305	0.2745	-0.6190	-3.39
2.310	0.2727	-0.6190	-3.40
2.315	0.2708	-0.6189	-3.41
2.320	0.2689	-0.6189	-3.42
2.325	0.2671	-0.6188	-3.43
2.330	0.2653	-0.6186	-3.44
2.335	0.2635	-0.6187	-3.45
2.340	0.2618	-0.6186	-3.46
2.345	0.2598	-0.6185	-3.47
2.350	0.2582	-0.6184	-3.48
2.355	0.2566	-0.6182	-3.49
2.360	0.2550	-0.6181	-3.50
2.365	0.2532	-0.6179	-3.51
2.370	0.2517	-0.6179	-3.52
2.375	0.2500	-0.6177	-3.53
2.380	0.2485	-0.6175	-3.54
2.385	0.2470	-0.6174	-3.54
2.390	0.2455	-0.6173	-3.55
2.395	0.2439	-0.6173	-3.56
2.400	0.2424	-0.6172	-3.57
2.405	0.2410	-0.6171	-3.58
2.410	0.2397	-0.6168	-3.59
2.415	0.2382	-0.6168	-3.59

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.420	0.2368	-0.6165	-3.60
2.425	0.2354	-0.6164	-3.61
2.430	0.2341	-0.6162	-3.62
2.435	0.2328	-0.6160	-3.63
2.440	0.2315	-0.6157	-3.64
2.445	0.2304	-0.6156	-3.64
2.450	0.2292	-0.6154	-3.65
2.455	0.2282	-0.6150	-3.66
2.460	0.2271	-0.6149	-3.67
2.465	0.2260	-0.6146	-3.68
2.470	0.2249	-0.6144	-3.69
2.475	0.2238	-0.6141	-3.69
2.480	0.2227	-0.6138	-3.70
2.485	0.2217	-0.6136	-3.71
2.490	0.2206	-0.6134	-3.72
2.495	0.2196	-0.6132	-3.72
2.500	0.2186	-0.6130	-3.73
2.505	0.2175	-0.6129	-3.74
2.510	0.2166	-0.6128	-3.74
2.515	0.2156	-0.6126	-3.75
2.520	0.2146	-0.6124	-3.76
2.525	0.2137	-0.6122	-3.76
2.530	0.2127	-0.6121	-3.77
2.535	0.2118	-0.6120	-3.77
2.540	0.2108	-0.6120	-3.78
2.545	0.2099	-0.6116	-3.79
2.550	0.2091	-0.6115	-3.79
2.555	0.2083	-0.6112	-3.80
2.560	0.2075	-0.6109	-3.81
2.565	0.2070	-0.6103	-3.82
2.570	0.2064	-0.6098	-3.83
2.575	0.2062	-0.6093	-3.83
2.580	0.2058	-0.6090	-3.84
2.585	0.2057	-0.6087	-3.84
2.590	0.2054	-0.6085	-3.85
2.595	0.2052	-0.6084	-3.85
2.600	0.2049	-0.6084	-3.85
2.605	0.2045	-0.6084	-3.85
2.610	0.2042	-0.6083	-3.85
2.615	0.2038	-0.6081	-3.86
2.620	0.2035	-0.6080	-3.86
2.625	0.2031	-0.6079	-3.86
2.630	0.2027	-0.6077	-3.87
2.635	0.2023	-0.6075	-3.87
2.640	0.2018	-0.6074	-3.88
2.645	0.2013	-0.6071	-3.88

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.650	0.2010	-0.6070	-3.88
2.655	0.2006	-0.6068	-3.89
2.660	0.2000	-0.6066	-3.89
2.665	0.1995	-0.6064	-3.90
2.670	0.1990	-0.6062	-3.90
2.675	0.1984	-0.6060	-3.91
2.680	0.1979	-0.6061	-3.91
2.685	0.1973	-0.6059	-3.91
2.690	0.1968	-0.6058	-3.92
2.695	0.1963	-0.6057	-3.92
2.700	0.1960	-0.6056	-3.92
2.705	0.1955	-0.6055	-3.93
2.710	0.1951	-0.6051	-3.93
2.715	0.1948	-0.6050	-3.94
2.720	0.1943	-0.6047	-3.94
2.725	0.1939	-0.6041	-3.95
2.730	0.1934	-0.6037	-3.96
2.735	0.1929	-0.6031	-3.97
2.740	0.1923	-0.6025	-3.98
2.745	0.1916	-0.6020	-3.99
2.750	0.1910	-0.6016	-4.00
2.755	0.1903	-0.6010	-4.01
2.760	0.1894	-0.6005	-4.02
2.765	0.1888	-0.5999	-4.03
2.770	0.1881	-0.5994	-4.04
2.775	0.1871	-0.5988	-4.05
2.780	0.1863	-0.5983	-4.06
2.785	0.1854	-0.5979	-4.07
2.790	0.1845	-0.5977	-4.08
2.795	0.1836	-0.5971	-4.09
2.800	0.1827	-0.5967	-4.10
2.805	0.1817	-0.5964	-4.10
2.810	0.1808	-0.5959	-4.11
2.815	0.1797	-0.5956	-4.12
2.820	0.1788	-0.5954	-4.13
2.825	0.1779	-0.5950	-4.14
2.830	0.1770	-0.5946	-4.15
2.835	0.1760	-0.5944	-4.15
2.840	0.1751	-0.5940	-4.16
2.845	0.1740	-0.5938	-4.17
2.850	0.1731	-0.5935	-4.18
2.855	0.1722	-0.5931	-4.19
2.860	0.1714	-0.5929	-4.19
2.865	0.1707	-0.5925	-4.20
2.870	0.1701	-0.5920	-4.21
2.875	0.1693	-0.5917	-4.22

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.880	0.1685	-0.5909	-4.23
2.885	0.1679	-0.5903	-4.24
2.890	0.1670	-0.5897	-4.25
2.895	0.1660	-0.5889	-4.27
2.900	0.1647	-0.5881	-4.28
2.905	0.1637	-0.5875	-4.30
2.910	0.1626	-0.5868	-4.31
2.915	0.1612	-0.5864	-4.32
2.920	0.1599	-0.5860	-4.33
2.925	0.1585	-0.5857	-4.34
2.930	0.1573	-0.5852	-4.35
2.935	0.1562	-0.5851	-4.36
2.940	0.1551	-0.5849	-4.36
2.945	0.1539	-0.5849	-4.37
2.950	0.1530	-0.5848	-4.37
2.955	0.1520	-0.5848	-4.38
2.960	0.1513	-0.5848	-4.38
2.965	0.1504	-0.5849	-4.38
2.970	0.1500	-0.5850	-4.38
2.975	0.1493	-0.5850	-4.38
2.980	0.1486	-0.5850	-4.38
2.985	0.1482	-0.5851	-4.39
2.990	0.1480	-0.5850	-4.39
2.995	0.1477	-0.5849	-4.39
3.000	0.1474	-0.5845	-4.40
3.005	0.1447	-0.5838	-4.42
3.010	0.1442	-0.5840	-4.41
3.015	0.1441	-0.5840	-4.42
3.020	0.1440	-0.5838	-4.42
3.025	0.1439	-0.5840	-4.42
3.030	0.1439	-0.5838	-4.42
3.035	0.1439	-0.5834	-4.42
3.040	0.1440	-0.5829	-4.43
3.045	0.1438	-0.5823	-4.44
3.050	0.1439	-0.5815	-4.45
3.055	0.1439	-0.5809	-4.46
3.060	0.1435	-0.5800	-4.47
3.065	0.1432	-0.5789	-4.49
3.070	0.1425	-0.5781	-4.50
3.075	0.1419	-0.5771	-4.52
3.080	0.1413	-0.5763	-4.53
3.085	0.1406	-0.5755	-4.55
3.090	0.1397	-0.5746	-4.56
3.095	0.1385	-0.5742	-4.57
3.100	0.1377	-0.5739	-4.58
3.105	0.1367	-0.5736	-4.59

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.110	0.1353	-0.5733	-4.60
3.115	0.1344	-0.5732	-4.60
3.120	0.1335	-0.5733	-4.60
3.125	0.1326	-0.5736	-4.60
3.130	0.1319	-0.5737	-4.60
3.135	0.1312	-0.5741	-4.60
3.140	0.1307	-0.5745	-4.59
3.145	0.1303	-0.5750	-4.59
3.150	0.1298	-0.5755	-4.58
3.155	0.1296	-0.5757	-4.58
3.160	0.1294	-0.5760	-4.58
3.165	0.1295	-0.5764	-4.57
3.170	0.1295	-0.5766	-4.57
3.175	0.1299	-0.5768	-4.56
3.180	0.1299	-0.5769	-4.56
3.185	0.1300	-0.5766	-4.57
3.190	0.1303	-0.5761	-4.57
3.195	0.1306	-0.5757	-4.58
3.200	0.1305	-0.5749	-4.59
3.205	0.1305	-0.5741	-4.60
3.210	0.1304	-0.5733	-4.61
3.215	0.1303	-0.5725	-4.63
3.220	0.1295	-0.5715	-4.64
3.225	0.1286	-0.5708	-4.66
3.230	0.1278	-0.5705	-4.66
3.235	0.1268	-0.5704	-4.67
3.240	0.1262	-0.5704	-4.67
3.245	0.1256	-0.5705	-4.67
3.250	0.1248	-0.5705	-4.67
3.255	0.1243	-0.5705	-4.67
3.260	0.1237	-0.5710	-4.67
3.265	0.1233	-0.5710	-4.67
3.270	0.1229	-0.5711	-4.67
3.275	0.1223	-0.5713	-4.67
3.280	0.1219	-0.5713	-4.67
3.285	0.1215	-0.5716	-4.67
3.290	0.1210	-0.5716	-4.67
3.295	0.1206	-0.5718	-4.67
3.300	0.1203	-0.5717	-4.67
3.305	0.1200	-0.5717	-4.67
3.310	0.1197	-0.5716	-4.67
3.315	0.1196	-0.5715	-4.67
3.320	0.1193	-0.5715	-4.67
3.325	0.1187	-0.5716	-4.68
3.330	0.1187	-0.5712	-4.68
3.335	0.1185	-0.5707	-4.69

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.340	0.1183	-0.5705	-4.69
3.345	0.1181	-0.5702	-4.70
3.350	0.1178	-0.5697	-4.70
3.355	0.1177	-0.5692	-4.71
3.360	0.1174	-0.5687	-4.72
3.365	0.1171	-0.5680	-4.73
3.370	0.1165	-0.5673	-4.74
3.375	0.1164	-0.5664	-4.76
3.380	0.1157	-0.5656	-4.77
3.385	0.1150	-0.5648	-4.79
3.390	0.1146	-0.5641	-4.80
3.395	0.1140	-0.5631	-4.81
3.400	0.1132	-0.5621	-4.83
3.405	0.1124	-0.5611	-4.85
3.410	0.1117	-0.5599	-4.87
3.415	0.1107	-0.5593	-4.88
3.420	0.1097	-0.5586	-4.89
3.425	0.1086	-0.5578	-4.91
3.430	0.1076	-0.5567	-4.93
3.435	0.1064	-0.5561	-4.94
3.440	0.1053	-0.5553	-4.96
3.445	0.1041	-0.5543	-4.97
3.450	0.1027	-0.5539	-4.99
3.455	0.1014	-0.5533	-5.00
3.460	0.1002	-0.5529	-5.01
3.465	0.0989	-0.5523	-5.02
3.470	0.0976	-0.5523	-5.02
3.475	0.0965	-0.5520	-5.03
3.480	0.0953	-0.5518	-5.04
3.485	0.0942	-0.5520	-5.04
3.490	0.0932	-0.5525	-5.03
3.495	0.0928	-0.5522	-5.04
3.500	0.0920	-0.5526	-5.03
3.505	0.0918	-0.5526	-5.03
3.510	0.0914	-0.5526	-5.03
3.515	0.0914	-0.5523	-5.04
3.520	0.0914	-0.5520	-5.04
3.525	0.0914	-0.5517	-5.05
3.530	0.0916	-0.5510	-5.06
3.535	0.0926	-0.5498	-5.07
3.540	0.0929	-0.5484	-5.10
3.545	0.0927	-0.5466	-5.12
3.550	0.0925	-0.5444	-5.16
3.555	0.0919	-0.5420	-5.20
3.560	0.0909	-0.5397	-5.24
3.565	0.0896	-0.5374	-5.28

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.570	0.0874	-0.5352	-5.32
3.575	0.0853	-0.5323	-5.37
3.580	0.0819	-0.5312	-5.39
3.585	0.0786	-0.5304	-5.41
3.590	0.0753	-0.5311	-5.41
3.595	0.0725	-0.5320	-5.40
3.600	0.0706	-0.5337	-5.38
3.605	0.0691	-0.5355	-5.35
3.610	0.0688	-0.5371	-5.33
3.615	0.0692	-0.5385	-5.30
3.620	0.0699	-0.5396	-5.29
3.625	0.0708	-0.5404	-5.27
3.630	0.0721	-0.5401	-5.27
3.635	0.0733	-0.5394	-5.28
3.640	0.0738	-0.5381	-5.30
3.645	0.0741	-0.5368	-5.32
3.650	0.0743	-0.5357	-5.34
3.655	0.0744	-0.5342	-5.36
3.660	0.0741	-0.5329	-5.38
3.665	0.0735	-0.5318	-5.40
3.670	0.0732	-0.5303	-5.43
3.675	0.0730	-0.5288	-5.45
3.680	0.0726	-0.5276	-5.47
3.685	0.0718	-0.5261	-5.50
3.690	0.0709	-0.5248	-5.52
3.695	0.0696	-0.5232	-5.55
3.700	0.0681	-0.5215	-5.58
3.705	0.0664	-0.5200	-5.61
3.710	0.0643	-0.5189	-5.63
3.715	0.0618	-0.5179	-5.65
3.720	0.0588	-0.5174	-5.67
3.725	0.0560	-0.5177	-5.67
3.730	0.0518	-0.5180	-5.67
3.735	0.0494	-0.5199	-5.64
3.740	0.0471	-0.5224	-5.60
3.745	0.0463	-0.5252	-5.56
3.750	0.0459	-0.5284	-5.51
3.755	0.0470	-0.5312	-5.46
3.760	0.0492	-0.5332	-5.43
3.765	0.0520	-0.5341	-5.41
3.770	0.0554	-0.5337	-5.41
3.775	0.0577	-0.5318	-5.43
3.780	0.0602	-0.5293	-5.47
3.785	0.0603	-0.5256	-5.53
3.790	0.0593	-0.5227	-5.58
3.795	0.0574	-0.5203	-5.62

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.800	0.0555	-0.5186	-5.65
3.805	0.0533	-0.5176	-5.67
3.810	0.0514	-0.5170	-5.69
3.815	0.0494	-0.5166	-5.70
3.820	0.0475	-0.5164	-5.70
3.825	0.0457	-0.5167	-5.70
3.830	0.0440	-0.5170	-5.70
3.835	0.0435	-0.5201	-5.65
3.840	0.0427	-0.5201	-5.65
3.845	0.0421	-0.5201	-5.65
3.850	0.0414	-0.5200	-5.65
3.855	0.0407	-0.5198	-5.66
3.860	0.0401	-0.5196	-5.66
3.865	0.0396	-0.5193	-5.67
3.870	0.0386	-0.5191	-5.67
3.875	0.0379	-0.5190	-5.67
3.880	0.0371	-0.5187	-5.68
3.885	0.0362	-0.5184	-5.69
3.890	0.0353	-0.5183	-5.69
3.895	0.0345	-0.5179	-5.70
3.900	0.0335	-0.5178	-5.70
3.905	0.0325	-0.5173	-5.71
3.910	0.0315	-0.5172	-5.71
3.915	0.0304	-0.5169	-5.72
3.920	0.0292	-0.5167	-5.72
3.925	0.0282	-0.5164	-5.73
3.930	0.0274	-0.5160	-5.73
3.935	0.0262	-0.5159	-5.74
3.940	0.0252	-0.5155	-5.75
3.945	0.0242	-0.5152	-5.75
3.950	0.0229	-0.5149	-5.76
3.955	0.0218	-0.5145	-5.76
3.960	0.0207	-0.5141	-5.77
3.965	0.0196	-0.5137	-5.78
3.970	0.0184	-0.5133	-5.79
3.975	0.0172	-0.5126	-5.80
3.980	0.0161	-0.5123	-5.80
3.985	0.0147	-0.5117	-5.82
3.990	0.0137	-0.5113	-5.82
3.995	0.0125	-0.5109	-5.83
4.000	0.0112	-0.5105	-5.84
4.005	0.0100	-0.5099	-5.85
4.010	0.0086	-0.5095	-5.86
4.015	0.0075	-0.5089	-5.87
4.020	0.0063	-0.5084	-5.88
4.025	0.0053	-0.5081	-5.88

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.030	0.0040	-0.5075	-5.89
4.035	0.0027	-0.5069	-5.90
4.040	0.0014	-0.5064	-5.91
4.045	0.0002	-0.5056	-5.92
4.050	-0.0010	-0.5051	-5.93
4.055	-0.0024	-0.5045	-5.94
4.060	-0.0034	-0.5039	-5.95
4.065	-0.0046	-0.5035	-5.96
4.070	-0.0058	-0.5027	-5.97
4.075	-0.0069	-0.5023	-5.98
4.080	-0.0080	-0.5018	-5.99
4.085	-0.0092	-0.5012	-6.00
4.090	-0.0104	-0.5007	-6.01
4.095	-0.0115	-0.5001	-6.02
4.100	-0.0127	-0.4998	-6.02
4.105	-0.0139	-0.4993	-6.03
4.110	-0.0152	-0.4989	-6.04
4.115	-0.0162	-0.4986	-6.04
4.120	-0.0173	-0.4982	-6.05
4.125	-0.0187	-0.4979	-6.05
4.130	-0.0203	-0.4975	-6.06
4.135	-0.0215	-0.4971	-6.06
4.140	-0.0227	-0.4965	-6.07
4.145	-0.0242	-0.4957	-6.08
4.150	-0.0254	-0.4951	-6.09
4.155	-0.0263	-0.4945	-6.10
4.160	-0.0273	-0.4938	-6.12
4.165	-0.0281	-0.4932	-6.12
4.170	-0.0288	-0.4925	-6.14
4.175	-0.0295	-0.4922	-6.14
4.180	-0.0301	-0.4916	-6.15
4.185	-0.0304	-0.4910	-6.16
4.190	-0.0310	-0.4907	-6.17
4.195	-0.0315	-0.4905	-6.17
4.200	-0.0319	-0.4902	-6.17
4.205	-0.0324	-0.4902	-6.17
4.210	-0.0328	-0.4899	-6.18
4.215	-0.0332	-0.4899	-6.18
4.220	-0.0334	-0.4896	-6.18
4.225	-0.0341	-0.4896	-6.18
4.230	-0.0341	-0.4895	-6.18
4.235	-0.0344	-0.4895	-6.18
4.240	-0.0347	-0.4894	-6.19
4.245	-0.0350	-0.4895	-6.18
4.250	-0.0352	-0.4895	-6.18
4.255	-0.0353	-0.4896	-6.18

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.260	-0.0354	-0.4896	-6.18
4.265	-0.0357	-0.4896	-6.18
4.270	-0.0357	-0.4897	-6.18
4.275	-0.0358	-0.4898	-6.18
4.280	-0.0359	-0.4898	-6.18
4.285	-0.0360	-0.4899	-6.17
4.290	-0.0359	-0.4899	-6.17
4.295	-0.0360	-0.4900	-6.17
4.300	-0.0361	-0.4900	-6.17
4.305	-0.0362	-0.4900	-6.17
4.310	-0.0362	-0.4901	-6.17
4.315	-0.0363	-0.4901	-6.17
4.320	-0.0366	-0.4902	-6.17
4.325	-0.0366	-0.4904	-6.17
4.330	-0.0368	-0.4906	-6.16
4.335	-0.0368	-0.4907	-6.16
4.340	-0.0370	-0.4910	-6.15
4.345	-0.0370	-0.4913	-6.15
4.350	-0.0370	-0.4917	-6.14
4.355	-0.0370	-0.4921	-6.13
4.360	-0.0369	-0.4925	-6.13
4.365	-0.0368	-0.4929	-6.12
4.370	-0.0365	-0.4933	-6.11
4.375	-0.0364	-0.4937	-6.11
4.380	-0.0360	-0.4941	-6.10
4.385	-0.0355	-0.4946	-6.09
4.390	-0.0351	-0.4949	-6.09
4.395	-0.0347	-0.4952	-6.08
4.400	-0.0343	-0.4954	-6.08
4.405	-0.0338	-0.4956	-6.08
4.410	-0.0335	-0.4955	-6.08
4.415	-0.0331	-0.4955	-6.08
4.420	-0.0327	-0.4956	-6.08
4.425	-0.0321	-0.4956	-6.08
4.430	-0.0319	-0.4955	-6.08
4.435	-0.0318	-0.4953	-6.08
4.440	-0.0317	-0.4952	-6.09
4.445	-0.0314	-0.4952	-6.09
4.450	-0.0315	-0.4952	-6.09
4.455	-0.0312	-0.4953	-6.09
4.460	-0.0309	-0.4953	-6.09
4.465	-0.0308	-0.4956	-6.08
4.470	-0.0302	-0.4957	-6.08
4.475	-0.0298	-0.4962	-6.07
4.480	-0.0292	-0.4969	-6.06
4.485	-0.0288	-0.4976	-6.05

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.490	-0.0282	-0.4983	-6.04
4.495	-0.0280	-0.4990	-6.02
4.500	-0.0272	-0.4998	-6.01
4.505	-0.0269	-0.5002	-6.01
4.510	-0.0262	-0.5004	-6.00
4.515	-0.0256	-0.5007	-6.00
4.520	-0.0248	-0.5008	-6.00
4.525	-0.0244	-0.5003	-6.00
4.530	-0.0238	-0.5002	-6.01
4.535	-0.0234	-0.5000	-6.01
4.540	-0.0233	-0.4995	-6.02
4.545	-0.0233	-0.4990	-6.03
4.550	-0.0232	-0.4989	-6.03
4.555	-0.0232	-0.4986	-6.04
4.560	-0.0232	-0.4982	-6.04
4.565	-0.0231	-0.4979	-6.05
4.570	-0.0231	-0.4974	-6.06
4.575	-0.0232	-0.4969	-6.06
4.580	-0.0232	-0.4966	-6.07
4.585	-0.0232	-0.4965	-6.07
4.590	-0.0233	-0.4957	-6.09
4.595	-0.0236	-0.4953	-6.09
4.600	-0.0238	-0.4947	-6.10
4.605	-0.0241	-0.4943	-6.11
4.610	-0.0241	-0.4940	-6.12
4.615	-0.0246	-0.4936	-6.12
4.620	-0.0250	-0.4930	-6.13
4.625	-0.0252	-0.4924	-6.14
4.630	-0.0254	-0.4922	-6.15
4.635	-0.0255	-0.4915	-6.16
4.640	-0.0258	-0.4909	-6.17
4.645	-0.0261	-0.4904	-6.18
4.650	-0.0260	-0.4901	-6.18
4.655	-0.0262	-0.4895	-6.19
4.660	-0.0265	-0.4892	-6.20
4.665	-0.0264	-0.4889	-6.20
4.670	-0.0264	-0.4882	-6.22
4.675	-0.0262	-0.4878	-6.22
4.680	-0.0264	-0.4875	-6.23
4.685	-0.0263	-0.4871	-6.24
4.690	-0.0264	-0.4868	-6.24
4.695	-0.0261	-0.4864	-6.25
4.700	-0.0261	-0.4858	-6.26
4.705	-0.0262	-0.4852	-6.27
4.710	-0.0259	-0.4846	-6.28
4.715	-0.0261	-0.4841	-6.29

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.720	-0.0264	-0.4833	-6.30
4.725	-0.0266	-0.4827	-6.31
4.730	-0.0270	-0.4819	-6.33
4.735	-0.0274	-0.4810	-6.34
4.740	-0.0281	-0.4804	-6.35
4.745	-0.0290	-0.4796	-6.37
4.750	-0.0297	-0.4788	-6.38
4.755	-0.0305	-0.4781	-6.39
4.760	-0.0314	-0.4775	-6.40
4.765	-0.0320	-0.4771	-6.41
4.770	-0.0325	-0.4767	-6.41
4.775	-0.0329	-0.4762	-6.42
4.780	-0.0333	-0.4758	-6.43
4.785	-0.0336	-0.4754	-6.44
4.790	-0.0333	-0.4750	-6.44
4.795	-0.0334	-0.4744	-6.46
4.800	-0.0332	-0.4738	-6.47
4.805	-0.0329	-0.4735	-6.47
4.810	-0.0325	-0.4726	-6.49
4.815	-0.0324	-0.4722	-6.50
4.820	-0.0321	-0.4712	-6.52
4.825	-0.0318	-0.4707	-6.53
4.830	-0.0318	-0.4699	-6.54
4.835	-0.0316	-0.4695	-6.55
4.840	-0.0314	-0.4692	-6.55
4.845	-0.0315	-0.4688	-6.56
4.850	-0.0316	-0.4684	-6.57
4.855	-0.0314	-0.4685	-6.57
4.860	-0.0317	-0.4682	-6.57
4.865	-0.0319	-0.4680	-6.58
4.870	-0.0321	-0.4680	-6.58
4.875	-0.0322	-0.4682	-6.57
4.880	-0.0324	-0.4682	-6.57
4.885	-0.0325	-0.4684	-6.57
4.890	-0.0327	-0.4685	-6.56
4.895	-0.0330	-0.4687	-6.56
4.900	-0.0330	-0.4689	-6.56
4.905	-0.0333	-0.4690	-6.56
4.910	-0.0334	-0.4690	-6.56
4.915	-0.0336	-0.4693	-6.55
4.920	-0.0338	-0.4695	-6.55
4.925	-0.0340	-0.4696	-6.54
4.930	-0.0341	-0.4697	-6.54
4.935	-0.0344	-0.4700	-6.54
4.940	-0.0346	-0.4700	-6.53
4.945	-0.0349	-0.4703	-6.53

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.950	-0.0350	-0.4706	-6.52
4.955	-0.0353	-0.4706	-6.52
4.960	-0.0355	-0.4706	-6.52
4.965	-0.0357	-0.4708	-6.52
4.970	-0.0360	-0.4710	-6.52
4.975	-0.0362	-0.4710	-6.51
4.980	-0.0364	-0.4712	-6.51
4.985	-0.0365	-0.4712	-6.51
4.990	-0.0369	-0.4715	-6.50
4.995	-0.0369	-0.4716	-6.50
5.000	-0.0372	-0.4717	-6.50
5.005	-0.0373	-0.4718	-6.50
5.010	-0.0375	-0.4718	-6.50
5.015	-0.0375	-0.4717	-6.50
5.020	-0.0376	-0.4717	-6.50
5.025	-0.0374	-0.4716	-6.50
5.030	-0.0374	-0.4714	-6.51
5.035	-0.0370	-0.4714	-6.50
5.040	-0.0368	-0.4711	-6.51
5.045	-0.0363	-0.4712	-6.51
5.050	-0.0359	-0.4711	-6.51
5.055	-0.0356	-0.4710	-6.51
5.060	-0.0352	-0.4708	-6.52
5.065	-0.0352	-0.4711	-6.51
5.070	-0.0345	-0.4708	-6.52
5.075	-0.0346	-0.4709	-6.52
5.080	-0.0342	-0.4706	-6.52
5.085	-0.0345	-0.4705	-6.53
5.090	-0.0340	-0.4704	-6.53
5.095	-0.0344	-0.4701	-6.53
5.100	-0.0345	-0.4700	-6.54
5.105	-0.0346	-0.4698	-6.54
5.110	-0.0350	-0.4697	-6.54
5.115	-0.0352	-0.4693	-6.55
5.120	-0.0354	-0.4692	-6.55
5.125	-0.0356	-0.4688	-6.56
5.130	-0.0360	-0.4687	-6.56
5.135	-0.0361	-0.4682	-6.57
5.140	-0.0364	-0.4678	-6.57
5.145	-0.0366	-0.4677	-6.57
5.150	-0.0370	-0.4672	-6.58
5.155	-0.0374	-0.4666	-6.59
5.160	-0.0378	-0.4662	-6.60
5.165	-0.0381	-0.4655	-6.61
5.170	-0.0387	-0.4651	-6.62
5.175	-0.0392	-0.4643	-6.63

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.180	-0.0396	-0.4637	-6.64
5.185	-0.0402	-0.4631	-6.65
5.190	-0.0408	-0.4623	-6.67
5.195	-0.0414	-0.4614	-6.68
5.200	-0.0418	-0.4606	-6.70
5.205	-0.0424	-0.4598	-6.71
5.210	-0.0432	-0.4589	-6.73
5.215	-0.0438	-0.4581	-6.74
5.220	-0.0445	-0.4571	-6.76
5.225	-0.0450	-0.4561	-6.78
5.230	-0.0458	-0.4552	-6.79
5.235	-0.0465	-0.4543	-6.81
5.240	-0.0471	-0.4532	-6.83
5.245	-0.0479	-0.4522	-6.84
5.250	-0.0486	-0.4512	-6.86
5.255	-0.0493	-0.4501	-6.88
5.260	-0.0503	-0.4490	-6.90
5.265	-0.0511	-0.4480	-6.92
5.270	-0.0519	-0.4471	-6.93
5.275	-0.0526	-0.4462	-6.95
5.280	-0.0534	-0.4452	-6.97
5.285	-0.0543	-0.4443	-6.98
5.290	-0.0550	-0.4435	-7.00
5.295	-0.0559	-0.4427	-7.01
5.300	-0.0566	-0.4419	-7.02
5.305	-0.0575	-0.4411	-7.04
5.310	-0.0583	-0.4403	-7.05
5.315	-0.0589	-0.4395	-7.06
5.320	-0.0597	-0.4386	-7.08
5.325	-0.0604	-0.4378	-7.09
5.330	-0.0609	-0.4372	-7.10
5.335	-0.0615	-0.4364	-7.12
5.340	-0.0622	-0.4355	-7.13
5.345	-0.0627	-0.4348	-7.14
5.350	-0.0635	-0.4338	-7.16
5.355	-0.0642	-0.4328	-7.18
5.360	-0.0647	-0.4319	-7.20
5.365	-0.0655	-0.4309	-7.21
5.370	-0.0661	-0.4299	-7.23
5.375	-0.0671	-0.4290	-7.25
5.380	-0.0679	-0.4279	-7.27
5.385	-0.0690	-0.4271	-7.28
5.390	-0.0698	-0.4262	-7.29
5.395	-0.0708	-0.4254	-7.30
5.400	-0.0717	-0.4247	-7.32
5.405	-0.0727	-0.4241	-7.32

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.410	-0.0736	-0.4235	-7.33
5.415	-0.0745	-0.4231	-7.34
5.420	-0.0754	-0.4222	-7.35
5.425	-0.0765	-0.4216	-7.36
5.430	-0.0772	-0.4210	-7.37
5.435	-0.0781	-0.4207	-7.37
5.440	-0.0788	-0.4202	-7.38
5.445	-0.0797	-0.4197	-7.39
5.450	-0.0803	-0.4192	-7.40
5.455	-0.0810	-0.4186	-7.40
5.460	-0.0816	-0.4185	-7.40
5.465	-0.0822	-0.4177	-7.42
5.470	-0.0828	-0.4174	-7.42
5.475	-0.0835	-0.4167	-7.43
5.480	-0.0843	-0.4162	-7.44
5.485	-0.0852	-0.4156	-7.45
5.490	-0.0860	-0.4150	-7.46
5.495	-0.0868	-0.4146	-7.46
5.500	-0.0878	-0.4142	-7.47
5.505	-0.0886	-0.4137	-7.47
5.510	-0.0897	-0.4137	-7.47
5.515	-0.0905	-0.4134	-7.47
5.520	-0.0914	-0.4135	-7.46
5.525	-0.0920	-0.4135	-7.46
5.530	-0.0929	-0.4136	-7.46
5.535	-0.0933	-0.4135	-7.45
5.540	-0.0941	-0.4136	-7.45
5.545	-0.0945	-0.4136	-7.45
5.550	-0.0952	-0.4136	-7.44
5.555	-0.0958	-0.4137	-7.44
5.560	-0.0963	-0.4137	-7.44
5.565	-0.0969	-0.4138	-7.43
5.570	-0.0976	-0.4137	-7.43
5.575	-0.0982	-0.4138	-7.43
5.580	-0.0988	-0.4138	-7.42
5.585	-0.0996	-0.4137	-7.42
5.590	-0.1004	-0.4138	-7.42
5.595	-0.1012	-0.4139	-7.41
5.600	-0.1021	-0.4140	-7.40
5.605	-0.1029	-0.4143	-7.39
5.610	-0.1036	-0.4146	-7.38
5.615	-0.1045	-0.4151	-7.37
5.620	-0.1051	-0.4154	-7.36
5.625	-0.1058	-0.4158	-7.35
5.630	-0.1063	-0.4162	-7.34
5.635	-0.1068	-0.4168	-7.33

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.640	-0.1073	-0.4172	-7.32
5.645	-0.1076	-0.4176	-7.31
5.650	-0.1080	-0.4179	-7.30
5.655	-0.1084	-0.4182	-7.29
5.660	-0.1088	-0.4184	-7.28
5.665	-0.1091	-0.4187	-7.28
5.670	-0.1096	-0.4191	-7.27
5.675	-0.1099	-0.4194	-7.26
5.680	-0.1104	-0.4197	-7.25
5.685	-0.1106	-0.4198	-7.25
5.690	-0.1108	-0.4200	-7.24
5.695	-0.1111	-0.4202	-7.24
5.700	-0.1114	-0.4205	-7.23
5.705	-0.1115	-0.4208	-7.22
5.710	-0.1117	-0.4209	-7.22
5.715	-0.1120	-0.4212	-7.21
5.720	-0.1120	-0.4211	-7.22
5.725	-0.1123	-0.4214	-7.21
5.730	-0.1125	-0.4215	-7.21
5.735	-0.1125	-0.4216	-7.20
5.740	-0.1128	-0.4217	-7.20
5.745	-0.1129	-0.4221	-7.19
5.750	-0.1130	-0.4217	-7.20
5.755	-0.1130	-0.4217	-7.20
5.760	-0.1130	-0.4217	-7.20
5.765	-0.1131	-0.4215	-7.20
5.770	-0.1133	-0.4213	-7.20
5.775	-0.1132	-0.4212	-7.21
5.780	-0.1135	-0.4208	-7.21
5.785	-0.1135	-0.4206	-7.22
5.790	-0.1137	-0.4201	-7.23
5.795	-0.1138	-0.4197	-7.23
5.800	-0.1136	-0.4193	-7.24
5.805	-0.1139	-0.4189	-7.25
5.810	-0.1139	-0.4183	-7.26
5.815	-0.1140	-0.4179	-7.27
5.820	-0.1142	-0.4172	-7.28
5.825	-0.1146	-0.4166	-7.29
5.830	-0.1147	-0.4159	-7.30
5.835	-0.1148	-0.4156	-7.31
5.840	-0.1150	-0.4147	-7.32
5.845	-0.1150	-0.4141	-7.34
5.850	-0.1153	-0.4134	-7.35
5.855	-0.1158	-0.4125	-7.36
5.860	-0.1160	-0.4117	-7.38
5.865	-0.1164	-0.4112	-7.38

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.870	-0.1169	-0.4103	-7.40
5.875	-0.1172	-0.4097	-7.41
5.880	-0.1178	-0.4089	-7.42
5.885	-0.1182	-0.4083	-7.43
5.890	-0.1185	-0.4076	-7.44
5.895	-0.1189	-0.4072	-7.45
5.900	-0.1193	-0.4066	-7.46
5.905	-0.1199	-0.4059	-7.47
5.910	-0.1201	-0.4055	-7.47
5.915	-0.1203	-0.4051	-7.48
5.920	-0.1206	-0.4045	-7.49
5.925	-0.1210	-0.4039	-7.50
5.930	-0.1213	-0.4033	-7.51
5.935	-0.1214	-0.4026	-7.52
5.940	-0.1218	-0.4017	-7.54
5.945	-0.1223	-0.4011	-7.55
5.950	-0.1225	-0.4003	-7.56
5.955	-0.1230	-0.3994	-7.58
5.960	-0.1235	-0.3986	-7.59
5.965	-0.1241	-0.3978	-7.60
5.970	-0.1250	-0.3972	-7.61
5.975	-0.1259	-0.3967	-7.61
5.980	-0.1267	-0.3965	-7.61
5.985	-0.1273	-0.3963	-7.61
5.990	-0.1280	-0.3960	-7.62
5.995	-0.1287	-0.3960	-7.61
6.000	-0.1292	-0.3959	-7.61

2 Measurement Quality

Reliable band: from 45 MHz (S11 error σ = 1.0e-03); the points below it are kept in the plot (shaded, outside the scale) and in the tables, but are not reliable.
Spike removal: on (isolated point, 2nd difference > 6 x local noise and > S11 error budget, not shared with another sweep); replaced 1 point(s): Short at 4885 MHz ($|\Delta S11|$ = 1.4e-03).
Outliers shared by several sweeps, left alone (instrument artefacts cancelled by the calibration): 3835 MHz.

- Unreliable below 45 MHz: 8 point(s) (3 mm probe, S11 error σ = 1.0e-03). There the reference liquids resolve ε_r only to ±17.2 at 5 MHz … ±2.1 at 40 MHz per σ (usable: $\leq$ ±2). Values kept, but left out of the chart scale and of the summary medians; start the sweep at $\geq$ 45 MHz.

- Gn shows no radiation signature even at the top of the sweep (3 GHz – 6 GHz): mean arg(Gn) = -108° (radiation: -90°, median deviation 20°) and $|Gn| \propto f^{0.4}$ (radiation: f^3). Over the whole sweep it holds noise and the inconsistency between the reference liquids instead (its term is ≈7% of the Ref 1 (Distilled Water) admittance), so the full method's radiation correction is empirical here: compare with the simplified cross-check, especially for samples whose ε_r lies outside the range of the reference liquids.

- 2 frequency point(s) excluded: reference liquids nearly indistinguishable ($|S11_ref1 - S11_ref2| < 1e\text{-}02$). Consider starting the sweep above the affected range.